

SSC CGL (COMBINED GRADUATE LEVEL - INSPECTOR, ASO, TAX ASST) — SUBJECT COMPLETE QUESTION BANK

PRACTICE MATERIAL • VERIFIED QUESTION BANK

Total Questions: 444 • Mode: Single Subject Comprehensive Practice

- Q.1.** '©@'%@ 'YG 'éB')M'ò .y 20% 'Y@ "YC'iM'y? "TMK'%G '©0 •ù 'yC"TM#" "2 '©(" '©\$ 'éG•" "ù\$'%G '©M" \$"ù6'B " 'é@ 'Y0"
[English]: If the price of sugar increases by 20%, by what percent must a household reduce consumption so that expenditure remains unchanged?
A) 16.67% (16 !T%) B) 20%
C) 25% D) 15%
- Q.2.** •ù 'iA'Y>'%&"é0 'Y?'%@ "Y8"Ù\$" * " # R " R " &"2 "Ù0'é?'R ") (Successive Discounts) 'iG'I> "TMH-B 'Y2 "%.'IA")M'ò , 'ùM
[English]: A shopkeeper offers two successive discounts of 20% and 10%. What is the single equivalent discount?
A) 28% B) 30%
C) 25% D) 27%
- Q.3.** •ù 'ÉG•%."é("YM'ù>'©>" @ •Y*'%@ "Y8"Ù\$" • "2 "Ù0'ò .")2"Ù/ '©0 'ÉG'©("r "â &"é5"â " \$"â 9", * " 'IA 1 'Y?'yM" > 'YG "%M"
[English]: A dishonest dealer professes to sell at cost price but uses a false weight of 900g instead of 1kg. Find his profit percentage.
A) 11.11% (11 !Q%) B) 10%
C) 12.5% D) 9%
- Q.4.** 'YK•, "%0"é6"ò 8"é"é0'2 , "Ù/"é 'Y@ 'i0 "%G 8 "Y0"Ù7"1 'éG•" &"1 " (" 9"2 "é\$" 9"%d "Y9" "%0"é6"ò "ù\$'%G "Y0"Ù7"1 'éG•"
[English]: A sum doubles itself in 8 years at simple interest. In how many years will it become 4 times?
A) 24 "Y0"Ù7 / 24 Years B) 16 "Y0"Ù7 / 16 Years
C) 32 "Y0"Ù7 / 32 Years D) 20 "Y0"Ù7 / 20 Years
- Q.5.** 10,000 'Y@ " >"i? '©0 10% "Y>" M"y?'R &" 8"r " 5" M"r "r 'YM" 5"9&"Ù"ò , "Ù/"é 'i0 "%>'y>" # 'ÉM'ù>'Â "â •)\$ "F-f'erence
[English]: Find the difference between CI and SI on 10,000 at 10% per annum for 2 years:
A) 100 B) 150
C) 1200 D) 150
- Q.6.** 'ù&"ò B = 2 : 3 'l%"â " C = 4 : 5 "TMK, 'IK A : B : C 'Y> •Y(" *é\$ 'YM'ù> "TMK'y>?
[English]: If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$:
A) 8 : 12 : 15 B) 6 : 9 : 15
C) 8 : 10 : 15 D) 4 : 6 : 10
- Q.7.** A 'Y?'%@ 'Y>" M'ò "2 &"ù('éG•" \$'Y> B •TM8" "é0"Ù/ 'YK 15 'i?', .y '©B" > 'Y0 "%0 'i> "TMH-B &"1("1 'é?) " " , "é0"Ù/ 'YK 'Y"
[English]: A can do a piece of work in 10 days and B in 15 days. Working together, in how many days will they finish the work?
A) 6 'i?', ò b Fys B) 8 'i?', ò , Fys
C) 5 'i?', ò R Fys D) 7 'i?', ò r Fys
- Q.8.** '©>•* A •ù 'Y@ 'YK 6 '% 'ùG 'éG•" -" 8'Y\$â 9", " *é'¢ " "%G 8 '% 'ùG 'éG•" "é2" " 8'Y\$â 9"%d 'iK'%K•" *é'¢ 'R 8"é"
[English]: Pipe A fills a tank in 6 hours and Pipe B empties it in 8 hours. If both open together, in how many hours will the tank fill?
A) 24 '% 'ùG / 24 Hours B) 14 '% 'ùG / 14 Hours
C) 18 '% 'ùG / 18 Hours D) 12 '% 'ùG / 12 Hours
- Q.9.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'% 'ù> 'Y@ '©>" 8"r 'R •)-"r "2 "ù\$'%G "%G'Y ' .y '©>" " G'y@?
[English]: A 180m long train running at 54 km/h will cross a pole in how many seconds?
A) 12 "%G'Y ' ò " 6V6öæG0 B) 10 "%G'Y ' ò 6V6öæG0
C) 15 "%G'Y ' ò R 6V6öæG0 D) 18 "%G'Y ' ò , 6V6öæG0
- Q.10.** "i>•)\$ 'É2 'éG•" 'R ("é5 'Y@ '©>" "ù." ù •) "â 9", \$'Y> 'y>" > 'Y@ '©>" "ù." ù •) "â 9"%d 'y>" > 'YG •Y(" ")2 (Downstream)
[English]: Speed of a boat in still water is 10 km/h and stream speed is 2 km/h. How much time will it take to go 36 km downstream?
A) 3 '% 'ùG / 3 Hours B) 4 '% 'ùG / 4 Hours
C) 3.6 '% 'ùG / 3.6 Hours D) 4.5 '% 'ùG / 4.5 Hours
- Q.11.** '©M" %â S *Ù0"é "9\$ "ù "% 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð
[English]: What is the average of first 50 natural numbers?
A) 25.5 B) 25
C) 26 D) 50

- Q.12.** '©?'>'I0 '©A'IM" " 5" M'I."é(•/)" "â '%A'©>'B R 2 "™H–B B 5" M"r , "é& •™('Y@ •/)" "â /"1 43 "Y0"Ù7 "™K'y>–B *""ù\$ "â " [English]: The ratio of present ages of father and son is 5 : 2. After 4 years, sum of their ages will be 43. Find father's present age:
- A) 25 "Y0"Ù7 / 25 Years B) 30 "Y0"Ù7 / 30 Years
C) 35 "Y0"Ù7 / 35 Years D) 20 "Y0"Ù7 / 20 Years
- Q.13.** 'ù&"ò 8•) "Ù/"â C f ³cs2 ("Â f" 8"r *)0"Ù#I "Y?"Ù>ÉM'ò 9"²Â \$"² v²r "r 8"Ù%"é('©0 'YL', 8"â •) •i 'y>? [English]: If number 481k673 is completely divisible by 9, find the digit in place of 'k':
- A) 7 B) 2
C) 5 D) 6
- Q.14.** 'yA'9('12 (7 y u - 3 u x) 'éG•" 'Y>•, "â •) (Unit Digit) 'YM'ù> "™K'y>? [English]: Find the unit digit in the expression (7 y u - 3 u x):
- A) 4 B) 6
C) 3 D) 7
- Q.15.** 'iK "%o 'iM'ù>'9 'Y>•Y(" ""é\$ 3 : 4 "™H 'I%"â "%o "â .–i8–b ,,,4b' B 9"%od •™('Y> ")f"%of (LCM) 'YM'ù> "™K'y>? [English]: The ratio of two numbers is 3 : 4 and their HCF is 4. What is their LCM?
- A) 48 B) 36
C) 24 D) 60
- Q.16.** ¹40 '©M" \$"ò "ù "Ù0"â 5"é2"r "é5"" "² "c *""Ù0'I? 'Y?'yM" > "Y>")G '©>"Y2 'YG "%o>'R "ù8 •Y(" ""é\$ 'éG•" . "ù2"é/"â "é 'Y? 'é [English]: In what ratio must rice at ¹40/kg be mixed with rice at ¹60/kg so that mixture costs ¹45/kg?
- A) 3 : 1 B) 1 : 3
C) 2 : 1 D) 3 : 2
- Q.17.** •ù "YC'IM'B " "" ?'y? 44 "%oG'é@ "™H–B "%o "â "Ù7"y\$"Ù0'12 (Area) 'YM'ù> "™K'y>? [English]: The circumference of a circle is 44 cm. What is its area?
- A) 154 "%oG'é@² / 154 cm² B) 308 "%oG'é@² / 308 cm²
C) 77 "%oG'é@² / 77 cm² D) 616 "%oG'é@² / 616 cm²
- Q.18.** •ù " ' 'Â 5"9\$"Ù\$" / "i 'YA 'YG •i""é0 'Y@ 'IM" ?'ÉM'ù> 7 "%oG'é@ 'I%"â •) "é 24 "%oG'é@ "™H–B 6•) " "â 5'YM" *""97"Ù " / 'YM [English]: Base radius of a cone is 7 cm and height is 24 cm. Find its curved surface area:
- A) 550 "%oG'é@² / 550 cm² B) 500 "%oG'é@² / 500 cm²
C) 616 "%oG'é@² / 616 cm² D) 450 "%oG'é@² / 450 cm²
- Q.19.** A 'I0 B '%oG 'YM" . "i ¹20,000 'I0 ¹30,000 ") "é " 'R 5"Ù/"é*""é0 "iA" B 'Y?'ù>–B 5" M"r "r •)\$ 'éG•" " 2 ¹15,000 'Y>")>'Ò 9" – [English]: A and B started a business investing ¹20,000 and ¹30,000 respectively. Total annual profit is ¹15,000. What is B's share?
- A) ¹9,000 B) ¹6,000
C) ¹7,500 D) ¹10,000
- Q.20.** 'ù&"ò , ² ÷, Ò B 9"²Â \$"² <" ² ÷<" "â . "é('YM'ù> "™K'y>? [English]: If $x + 1/x = 4$, find the value of $x^2 + 1/x^2$:
- A) 14 B) 16
C) 18 D) 12
- Q.21.** 'ù&"ò F Â , = 4/3 "™K, 'IK sin ;, "â . "é('YM'ù> "™K'y>? [English]: If $\tan ;, \hat{O} B \hat{O} A \hat{v} \dagger B \text{---} 2 F \dagger R$ alue of $\sin ; f \hat{d}$
- A) 4/5 B) 3/5
C) 5/4 D) 3/4
- Q.22.** "YM'ù> 'É 24 ÷ 4 × (3 + 1) - 6 + 2 'Y> "%o0")\$'â . "é('YM'ù> "™K'y>? [English]: Simplify the expression $24 \div 4 \times (3 + 1) - 6 + 2$:
- A) 20 B) 18
C) 24 D) 16
- Q.23.** 'iK "YM'ù> "Ù\$"ò 'R 9" 8"Ù%"é("%oG 'YM" . "i 4 'Y?'é@/'%o 'ù> 'I0 5 'Y?'é@/'%o 'ù> 'Y@ 'y\$"ò 8"r 5"ù*" @'B &"ù6"é •" . "y '©2'I0 [English]: Two persons start from the same point in opposite directions at 4 km/h and 5 km/h. After 3 hours, distance between them is:
- A) 27 'Y?'é@ / 27 km B) 15 'Y?'é@ / 15 km
C) 3 'Y?'é@ / 3 km D) 30 'Y?'é@ / 30 km

- Q.24.** •ù '©A'%>"R .y 'iK •™."Ù." &"Y>" %yd 'É@'I("r 5"é2"r 'ém'é@'i5"é0 'YK 'YA"" 5"% 'é\$"'1 'Y> 60% 'é\$ 'é?")> 'I0 "Y9 1200 'é
[English]: In an election between two candidates, the winner got 60% of valid votes and won by 1200 votes. Total valid votes were:
A) 6,000 B) 5,000
C) 7,200 D) 4,800
- Q.25.** A 'I0 B 'Y?"%@ 'Y>" M'ò "2 "Ù0'é6•2 b &"ù('I0 8 'i?', .y 'Y0 "%@ 'IG "™H•)d '©B" G 'Y>" M'ò " 'É&")0" " C 9"%d A 'Y> "™H
[English]: A and B can do a work in 6 days and 8 days respectively. Total wages for the work is ₹1400. Find A's share:
A) ₹800 B) ₹600
C) ₹700 D) ₹750
- Q.26.** 'ù&"ò 'R 5"9\$"Ù\$ 'I0 •ù "Y0"Ù 'Y> '©0"ù."é* "%."é("™K, 'IK •™('YG 'YM"yG'IM" +")K•" "â '%A'©>'B "Ù/"â 9"1 "ãð
[English]: If the perimeter of a circle and a square are equal, what is the ratio of their areas?
A) 14 : 11 ("YC'IM'B "â, ' <"â' ò B 11 B) 11 : 14
C) 22 : 7 D) 1 : 1
- Q.27.** 'Y?"%@ "%@ 'iM'ù> 'YK 56 "%G 'Ù>'r &"y("r ** 6"y7"12 29 '©M" >'©M'B 9"1\$ "â 9"%d •™8" 8•) "Ù/"â "2, 8"r -"é 'iG'%G '©0 "iG
[English]: A number when divided by 56 gives a remainder of 29. What will be the remainder when divided by 8?
A) 5 B) 3
C) 7 D) 1
- Q.28.** ₹500 'Y@ 'y(" >"i? '©0 2 "Y0"Ù7 'Y> "%>'y>" # 'ÉM'ù>Â "C 9"%d "Y>" M"y?'R, "Ù/"é 'i0 'YM'ù> "™H?
[English]: The simple interest on ₹500 for 2 years is ₹40. What is the annual rate of interest?
A) 4% B) 5%
C) 6% D) 3.5%
- Q.29.** •ù 'É2"Ù2"y,"é '%G •Y*"%@ 11"Y@•" **é0" .y 100 " ('É("é 'É?"%8"r "%@ "â "%\$ 5 " ('É""Â 'ù>-B 5" '©>" @ 'YG 'É>'b
[English]: A batsman scores 100 runs in his 11th innings, increasing his average by 5 runs. Find his new average:
A) 50 " (/ 50 Runs B) 45 " (/ 45 Runs
C) 55 " (/ 55 Runs D) 60 " (/ 60 Runs
- Q.30.** 9 'I0 25 'Y> 'é""Ù/"é(" **é\$ " „ÖV â &÷ ÷ tional) 'YM'ù> "™K'y>?
[English]: What is the mean proportional between 9 and 25?
A) 15 B) 17
C) 12 D) 20
- Q.31.** A 'Y?"%@ 'Y>" M'ò "2 " &"ù("1 'éG•" " " "%@ 'Y>" M'ò "2 , &"ù("1 'éG•" **)0"â " 8'Y\$ "â 9"%d "YG 4 'i?'%K•" \$'R 'R 8"é0
[English]: A can do a work in 12 days and B in 18 days. They work together for 4 days, then A leaves. In how many days will B alone complete the remaining work?
A) 6 'i?'€ B) 8 'i?', f, F ys)
C) 10 'i?'€ D) 7 'i?'€
- Q.32.** 180 'é@'ù0 ") 'É@ •ù " G") "é!"É@ 54 'Y?'é@/'% "ù> 'Y@ 'y\$ "ò 8"r "" 0"™@ "™H-B 'R •)-r "2 **é0 'Y0'%G 'éG•" /"" "ù\$'%
[English]: A 180m long train is running at a speed of 54 km/h. How much time will it take to cross a pole?
A) 10 "%G'Y ' B) 12 "%G'Y ' f " 6V6ðæG2•
C) 15 "%G'Y ' D) 18 "%G'Y '
- Q.33.** ₹15,000 'Y@ 'y(" >"i? '©0 2 "Y0"Ù7 'YG ")?•ò R 5"é0"Ù7"ù 'i0 "%G '© "Ù0"YC'iM'y? 'ÉM'ù>Â " 8"é""é0'2, "Ù/"é 'Y> •Y 'I0 'Y
[English]: What is the difference between CI and SI on ₹15,000 for 2 years at 10% per annum?
A) ₹120 B) ₹150 (₹150)
C) ₹180 D) ₹200
- Q.34.** •ù 'iA'Y>%"&"é0 'Y?"%@ "Y8"Ù\$" "r •) "ù\$ 'éB")M'ò * " # R " ") 'iG'%G 'YG 'É>'b -" # R "â 2"é- •Y0"Ù "ù\$ 'Y0'I> "™H-B /
[English]: A shopkeeper gives 20% discount on marked price and still gains 20%. If cost price is ₹500, what is the marked price?
A) ₹650 B) ₹700
C) ₹750 (₹750) D) ₹800
- Q.35.** •ù 'YH")G 'éG•" " Â S **%8"r " #R **%8"r "r 8"ù "Ù "r 2 4 : 5 'YG •Y(" **é\$ 'éG•" 9"% -B /'i? 'YH")G 'éG•" " 2 'y(" >"i? ₹170
[English]: A bag contains ₹1, 50p and 25p coins in the ratio 3 : 4 : 5. If total amount is ₹170, find the number of 50p coins:
A) 80 B) 100
C) 120 (120 coins) D) 150

- Q.36.** •ù ") 'Â 5"9\$ "Ù\$ " / 'ÉG") ('YG •i"é0 'Y@ 'IM" ?'ÉM'ù> 7 "%G'é@ 'I0 •© '©>•, 8"y." 9"%d •y8'Y> 'YA"" ""97"Ù " / 'YM"yG'IM" •
 [English]: Find the Total Surface Area of a right circular cylinder with radius 7cm and height 10cm:
 A) 648 "%G'é@² B) 748 "%G'é@² (748 cm²)
 C) 848 "%G'é@² D) 924 "%G'é@²
- Q.37.** 'ù&"ò 6-â , + cos ;, Ò !£" 6÷2 , 'IK cos ;, Ò 6-â , 'Y> 'é>', "Ù/"â 9"1 "ãð
 [English]: If $\sin ;, ^2 6\div 2 , = " 2 \cos ;, \hat{A} F\hat{+}V\hat{a} v\hat{+} B \longrightarrow F\hat{+}R \hat{a}lue \text{ of } \cos ;, \hat{O} 6-\hat{a} , ?$
 A) " 2 sin ;£ B) " 2 tan ;£
 C) sin ;£ D) 1
- Q.38.** 'ù&"ò 8'éA'©M'©/ A '©0 "% 'É 'r " 8"Ù5'IA")M'ò ...&VfÆP×ive), "%.'é?'B ...7-ÖÖWG ic) 'I%"â 8•) "Ù0"é.'R ...@ransitive) 'I@'%"
 [English]: English: If a relation R on set A is reflexive, symmetric and transitive, then what is R called?
 A) " ?'YM'B 8•,•)' / Empty Relation B) 'IA")M'ù\$"â 8•,•)' / Equivalence Relation
 C) '©M" "\$ù8'é."ù\$ "% 'É 'r ò çF'×7-ÖÖWG ic Relation D) "%>" M"Y\$"Ù0"ù "% 'É 'r ò Væ—`ersal Relation
- Q.39.** 'ù&"ò +") (f: R !' R, f(x) = 2x 'iM"Y>" > '©0"ù-"é7"ù\$ "™H, 'IK f 'Y?" , ""Ù0'Y>" "â +") ("™H?
 [English]: English: If the function f: R !' R is defined by f(x) = 2x, then what type of function is f?
 A) •ù "%o " \$'Y> •i "Ù "é&'R ò ðæRÖðæR æB òçFð „&-|V7F—e)B) 'É9" Ù 'R \$'Y> •i "Ù "é&'R ò Ö ày-one and Onto
 C) •ù "%o " "" 'IA •Y("é "Ù "é&'R ò ðæRÖðæR ut not Onto D) ', "\$2 'YH'Y@ ' , 9" '©M'1>'i / Neither One-one nor Onto
- Q.40.** 'éA'IM'ò ."é("i>i> (Principal Value Branch) 'ÉG•" 6-â »f ó"" "â ."é('YM'ù> "™K'y>?
 [English]: English: What is the principal value of $\sin \{^1(1/2)\}$?
 A) < ób f3 • B) < ó2 fç •
 C) < óB fC[• D) < ó" f" •
- Q.41.** tan {^1(1) + cos {^1(-1/2) + sin {^1(-1/2) 'Y> 'é>', "Ù/"â 9"1 "ãð
 [English]: English: What is the value of $\tan \{^1(1) + \cos \{^1(-1/2) + \sin \{^1(-1/2)\}$?
 A) < ó@ B) < ó
 C) 3< ó@ D) <
- Q.42.** sin {^1(sin(2< ó2" "â 8"™@ 'é>', "Ù/"â 9"1 "ãð
 [English]: English: What is the correct value of $\sin \{^1(\sin(2< ó2""ð$
 A) 2< ó0 B) < ó0
 C) -< ó0 D) 4< ó0
- Q.43.** 'ù&"ò "YM'ùB"" Ò ¶ ö-¥Ö 'R 8'é."ù\$ •i5"Ù"/)9 (Symmetric Matrix) "™H, 'IK 'YL', 8" 6" M'B 8'IM'ò 9"1 " ð
 [English]: English: If matrix A = [a_ij] is a symmetric matrix, which condition is true?
 A) a_ij = -a_ji B) a_ij = a_ji
 C) a_ij = 0 D) a_ij = 1
- Q.44.** 'ù&"ò " " 8'é>', "1 "ò "r 5" M'r "YM'ùB"" 9"%o , 'IK (AB)' 'Y> 'é>', "ù8'YG 'É0"é," 9"1\$â 9"fð
 [English]: English: If A and B are square matrices of same order, then (AB)' is equal to:
 A) A'B' B) B'A'
 C) AB D) BA
- Q.45.** 'ù&"ò 'R 2 r 2 "1 "ò "â 5"Ù/" \$"Ù "Ù0'é#" / (Invertible) •i5"Ù"/)9 "™H, 'IK |adj(A)| 'Y> 'é>', "ù8'YG 'É0"é," 9"1 "ãð
 [English]: English: If A is an invertible matrix of order 3 × 3, then |adj(A)| is equal to:
 A) |A| B) |A|²
 C) |A|³ D) 3|A|
- Q.46.** 'ù&"ò "ù8" 8"é0'9?'R " "1 'iK '© 'YM'I?'ù>• ...&ðws) •Y%"Y> 'iK "%M'I 'Ò „6ðÇVÖç2' 8" M"Y8'â „-FVçF-6 Â' 9"1 , 'IK "%o>" #
 [English]: English: If any two rows or columns of a determinant are identical, what is the value of the determinant?
 A) 1 B) 0 / Zero
 C) -1 D) •Y*" ?'é?'B ò -æf-æ—FP
- Q.47.** •ù "Y0"Ù •i5"Ù"/)9 A "YM'ùA'IM'YM" .9@'ò „-àvertible) 'Y9">'I> "™H 'ù&"ò " "y5"" /i?:
 [English]: English: A square matrix A is called invertible if and only if:
 A) |A| = 0 B) |A| "" 0 (•Y5"Ù/" \$"Ù "Ù0'é#" / "%o9" , Non-singular)
 C) A = A' D) A = -A'
- Q.48.** '12', ' Ò Æör±6-â , 'â , "r 8"é*y "Ù7 •Y5'Y2'Â †G'öG,' "Ù/"â 9"1 "ãð
 [English]: English: What is the derivative (dy/dx) of $y = \log(\sin x)$ with respect to x?
 A) tan x B) cot x
 C) -cot x D) sec x

- Q.49.** 'ù&"ò ' Ò F â »', f7, Ò <2' ò f Ò 7<"" 9"2Â \$"2 G'öG, "â . "é('YM'ù> "TMK'y>?
 [English]: English: If $y = \tan \{^1((3x - x^3) / (1 - 3x^2))\}$, then what is dy/dx ?
 A) $3 / (1 + x^2)$ B) $1 / (1 + x^2)$
 C) $3 / (1 - x^2)$ D) $2 / (1 + x^2)$
- Q.50.** 'ù&"ò , Ò 6÷2 , 'l%"â ' Ò 6-â , "TMK, 'IK dy/dx 'Y> 'é>', "Ù/"â 9"1 "ãð
 [English]: English: If $x = a \cos ;$, æB ' Ò 6-â , then dy/dx is equal to:
 A) $\tan ;\epsilon$ B) $-\cot ;\epsilon$
 C) $\cot ;\epsilon$ D) $-\tan ;\epsilon$
- Q.51.** "YC'IM'B "r "Ù7"y\$"Ù0'12 'éG•" \$"Ù0"ù "Ù/"â " "r 8"é*y "Ù7 '©0"ù5" M'I('Y@ 'i0 'YM'ù> "TMK'y@ 'É, r = 5 cm "TMK?
 [English]: English: What is the rate of change of area of a circle with respect to radius r when r = 5 cm?
 A) $10 < 6\sqrt{6}\pi$ B) $5 < 6\sqrt{6}\pi$
 C) $25 < 6\sqrt{6}\pi$ D) $20 < 6\sqrt{6}\pi$
- Q.52.** "Y "Ù0 $y = x^3 - x$ 'YG 'É?•)&" , Ò " * 8"Ù* M"b 0"y "â " *"Ù0"Y#I> (Slope of Tangent) 'YM'ù> "TMK'y@?
 [English]: English: What is the slope of the tangent to the curve $y = x^3 - x$ at the point $x = 2$?
 A) 11 B) 12
 C) 10 D) 6
- Q.53.** "+ (1 / (1 + x^2)) dx 'Y> 'é>', "Ù/"â 9"1\$ "â 9" fð
 [English]: English: What is the value of "+ (1 / (1 + x^2)) dx?
 A) $\sin \{^1 x + C$ B) $\tan \{^1 x + C$
 C) $\log(1 + x^2) + C$ D) $\cos \{^1 x + C$
- Q.54.** "+ sec x dx 'Y> "%."é "('é>', "Ù/"â 9"1 "ãð
 [English]: English: What is the integral value of "+ sec x dx?
 A) $\log |\sec x + \tan x| + C$ B) $\log |\sec x - \tan x| + C$
 C) $\sec x \tan x + C$ D) $\tan^2 x + C$
- Q.55.** "+ e.2 ±6-â , ^2 6÷2 , ' G, "â . "é('YM'ù> "TMK'y>?
 [English]: English: What is the value of "+ e.2 ±6-â , ^2 6÷2 , ' G fð
 A) $e.2 6÷2 , ^2 0$ B) $e.2 6-â , ^2 0$
 C) $-e.2 6-â , ^2 0$ D) $e.2 ±6-â , Ò 6÷2 , ^2 0$
- Q.56.** '%? "iM'©?'B 8'é>'Y2', "2 â f Â/2) (sin x / (sin x + cos x)) dx 'Y> 'é>', "Ù/"â 9"1 "ãð
 [English]: English: What is the value of the definite integral "+ $\epsilon^{< \acute{o}} \pm 6-â , Ò \pm 6-â , ^2 6÷2 ,$ " G fð
 A) < ó B) < ó@
 C) 1 D) 0
- Q.57.** 'ù&"ò b±, 'R 5"ù7'â +")((Odd Function, $f(-x) = -f(x)$) "TMH, 'IK "+ < • C f(x) dx 'Y> 'é>', "Ù/"â 9"1 "ãð
 [English]: English: If $f(x)$ is an odd function ($f(-x) = -f(x)$), what is the value of "+ < • C f(x) dx?
 A) $2 "+ \epsilon C f(x) dx$ B) 0 / Zero
 C) a D) 1
- Q.58.** •Y5'Y2 "%." " # (d^2y/dx^2)^3 + (dy/dx)^2 + sin(dy/dx) + 1 = 0 'Y@ '%>'B „FVpree) 'YM'ù> "TMH?
 [English]: English: What is the degree of the differential equation $(d^2y/dx^2)^3 + (dy/dx)^2 + \sin(dy/dx) + 1 = 0$?
 A) 3 B) 2
 C) 1 D) '©0"ù-"é7"ù\$ '%9" / Not Defined
- Q.59.** " H'i?'R "Y "" 8'é@'Y0'2 G'öG, ^2 ' Ò "â 8'é>'Y2', " #R „-çFVprating Factor - I.F.) 'YM'ù> "TMK'I> "TMH?
 [English]: English: What is the integrating factor (I.F.) of linear differential equation $dy/dx + Py = Q$?
 A) $e^{(+ P dx)}$ B) $e^{(+ Q dx)}$
 C) $"+ P dx$ D) $e^{(- "+ P dx)}$
- Q.60.** •Y5'Y2 "%." " # $dy/dx = y/x$ 'Y> "YM'ù>'© "TM2 (General Solution) 'YM'ù> "TMK'y>?
 [English]: English: What is the general solution of the differential equation $dy/dx = y/x$?
 A) $y = Cx$ B) $y = C / x$
 C) $y = x^2 + C$ D) $xy = C$
- Q.61.** 'ù&"ò &"2 6")("Ù"y\$ "Ù\$ 8'i?"iK•" r " "r "r 2"ù a × • b × = 0 "TMK, 'IK •TM('YG 'É@'¢ "â "1# , , "Ù/"â 9"1 "ãð
 [English]: English: If for two non-zero vectors a × and b ×, $a \times b \times = 0$, what is the angle ;, &WGpeen them?
 A) 0° B) 45°
 C) 90° (< ó"Â * 8"Ù* 2•), "Y\$ / Perpendicular) D) 180°

Q.62. $i0'' r \text{ ‡} \mathbb{E} \times k0'''$ "â . "é('YM'ù> "™K'y>?

[English]: English: What is the value of $i0'' r \text{ ‡} \mathbb{E} \times k0'''$?

- A) 0 B) 1
C) -1 D) $k0$

Q.63. $\%0\&"\text{ù}6 a \times = 2i0''^2 6\mathbb{E} + 6k0''$ "â * " ?'é>'2 „Ö væ—GVFR Æ wÂ' "ù\$'%0> "™K'y>?

[English]: English: What is the magnitude of vector $a \times = 2i0''^2 6\mathbb{E} + 6k0\#0$?

- A) 7 B) 11
C) 49 D) " 41

Q.64. 'ù&"ò r r " r Ò r 9"2Â \$"2 8'i?"b r " " r " r 8•),)' 'éG•" "Ù/"â 8'IM'ò 9"f ð

[English]: English: If $a \times \times b \times = 0 \times$, what is true regarding vectors $a \times$ and $b \times$?

- A) $a \times \perp b \times$ "©0"%M'©0 " 'É5'B 9"% / Perpendicular B) $a \times \parallel b \times$ "©0"%M'©0 "%0. "é("é 'l0 / "%0 " G'b 9"% (Parallel / Collinear)
C) $|a \times| = |b \times|$ D) $a \times + b \times = 0 \times$

Q.65. 'ù&"ò 'R 0"y "â " &"ù "ÒÙ "1 "Ù/"é • „F—&V7F—öâ 6÷6—æW2' ÂÂ ÒÂ â 9"1 , 'IK 'YL', 8"â 8•),)' "%0&"%05 "%0\$"Ù/ "™H?

[English]: English: If l, m, n are the direction cosines of a line, which relation is always true?

- A) $l + m + n = 1$ B) $l^2 + m^2 + n^2 = 1$
C) $l^2 + m^2 + n^2 = 0$ D) $l^2 - m^2 + n^2 = 1$

Q.66. 'É?•)&" ‡,Â •, z) 'Y@ 'éB"" „ù 'iA (Origin - 0, 0, 0) "%0G 'iB" @ 'YM'ù> "™K'y@?

[English]: English: What is the distance of point (x, y, z) from the origin?

- A) $x + y + z$ B) " $(x^2 + y^2 + z^2)$
C) $x^2 + y^2 + z^2$ D) " $(x + y + z)$

Q.67. 'iK " G'i>•ù 'É?'%0 "r &"ù "ÒÙ "%0A'©>'B "Ù0'é6•2 Â " Â 2 '\$'Y> a , b , c , "™H•"Â * 8"Û* 2•), "Y\$ (Perpendicular) "™K•) "

[English]: English: Two lines with direction ratios $a \bullet, b \bullet, c \bullet$ and $a \bullet, b \bullet, c \bullet$ are perpendicular if:

- A) $a \bullet/a \bullet = b \bullet/b \bullet = c \bullet/c \bullet$, B) $a \bullet/a \bullet + b \bullet/b \bullet + c \bullet/c \bullet = 0$
C) $a \bullet/a \bullet + b \bullet/b \bullet + c \bullet/c \bullet = 1$ D) $a \bullet/b \bullet - a \bullet/b \bullet = 0$

Q.68. " H'i?'R * "Ù0"1 "Ù0'é., 8'è8"Ù/"â „Â ' . "y •™&"Ù&"y6"Ù/ '12', „ð&V7F—'e Function) $Z = ax + by$ 'Y> •y7"Ù 'l. 'é>', „÷ F—Ö Â Val

[English]: English: In a Linear Programming Problem (LPP), where does the optimal value of the objective function $Z = ax + by$ lie?

- A) "%0A"%0 'y\$ 'YM'yG'IM" "r 6" 0"Ù7 ('YK'9@'ò' „ù 'iA'9 '©0 / Corner points or feasible region Only at origin
C) "%0A"%0 'y\$ 'YM'yG'IM" "r " 'YG•)&"Ù0 '©0 / Exact center D) 'YM'yG'IM" "r "ù8" -" / "é&9"Ù "ù 'É?•)&" * " ò ày random point

Q.69. 'ù&"ò " " &"2 8"Ù5'1 'IM" 'ù("é • „æFW VæFVçB W'ents) "™K•"Â \$"2 „ " " " "â . "é('YM'ù> "™K'y>?

[English]: English: If A and B are two independent events, then what is $P(A \cap B)$?

- A) $P(A) + P(B)$ B) $P(A) \times P(B)$
C) $P(A) / P(B)$ D) $P(A) - P(B)$

Q.70. 'ù&"ò „ ' Ò 2ó,Â „ " Ò ó" '\$'Y> $P(A \cap B) = 1/4$ "™K, 'iK $P(A | B) 'Y> 'é>$, "Ù/"â 9"1 "ã ð

[English]: English: If $P(A) = 3/8, P(B) = 1/2$ and $P(A \cap B) = 1/4$, what is $P(A | B)$?

- A) $1/2$ B) $2/3$
C) $3/4$ D) $1/8$

Q.71. 'Y?'%0@ '%0 '%0> E 'Y@ '©M" >'ù?'Y\$ "â „R' '\$'Y> •™8'Y@ '©B" '%0 '%0> $P(E) 'Y> 'ùK'r "Ù/"â 9"1$ "â 9" f ð$

[English]: English: What is the sum of probability of an event $P(E)$ and its complement event $P(E')$?

- A) 0 B) 1
C) 0.5 D) •Y("ù6"Ù "ù\$ / Indeterminate

Q.72. 'ù&"ò 'R 5" M'r "YM'ùB"" 9",Â \$"2 2 r 8'iH"R "ù8 '©M" "é0 'Y> •i5"Ù/"9 "™K'I> "™H?

[English]: English: If A is a square matrix, then $A + A'$ is always which type of matrix?

- A) "Y?"y.-"%0. 'é?'B "YM'ùB"" ò 6"ew-symmetric B) "%0. 'é?'B "YM'ùB"" ò 7—ÖÖWG ic Matrix
C) •y "é •i5"Ù/"9 / Identity Matrix D) "iB'%0M'ò "YM'ùB"" ò çVÆÂ Ö G ix

Q.73. '12', b‡, ' Ò <2 Ò 7, "â 8"Ù0% "é(/ •™ "Ù "ù7"Ù (Local Maxima) 'Y?" , "ù 'iA '©0 "™K'y>?

[English]: English: At which point does the function $f(x) = x^3 - 3x$ have a local maximum?

- A) $x = 1$ B) $x = -1$
C) $x = 0$ D) $x = 3$

Q.74. "+ (1 / " $(a^2 - x^2)) dx$ 'Y> 'é>', "Ù/"â 9"1\$ "â 9" f ð

[English]: English: What is the value of $\int (1 / " (a^2 - x^2)) dx$?

- A) $\sin \{1(x/a) + C$ B) $(1/a) \sin \{1(x/a) + C$
C) $\cos \{1(x/a) + C$ D) $\log|x + " (a^2 - x^2)| + C$

- Q.75.** 'i' 6•2 8'é>'Y2', ,—çFVpration by Parts) "%B'IM" "2 R b G, Ò R "2 b G, Ò "2 ·Rr , "2 b G, •Ò G, . "y '©M" %'â +")('©A'%"("r "â 8"™
[English]: English: In Integration by Parts, what is the standard rule for choosing the first function?
A) BODMAS '%?'ù. B) ILATE '%?'ù. (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential)
C) L'Hopital '%?'ù. D) Cramer '%?'ù.
- Q.76.** 2 tan {¹ x 'Y> 'é>', 6—â »" "r *iK•" . "y 'YM'ù> "™K'I> "™H?
[English]: English: What is the value of 2 tan {¹ x in terms of sin {¹ x?
A) sin {¹(2x / (1 + x²)) B) sin {¹(2x / (1 - x²))
C) sin {¹((1 - x²) / (1 + x²)) D) sin {¹(x / (1 + x²))
- Q.77.** 'ù&"ò "YM'ùB"" " "1 "ò " r 2 \$'Y> B 'Y@ 'YK'ù? 3 × 4 "™K, 'IK 'yA'9('12 •i5"Ù/")9 AB 'Y@ 'YK'ù? (Order) 'YM'ù> "™K'y@?
[English]: English: If matrix A is of order 2 × 3 and matrix B is of order 3 × 4, what is the order of matrix AB?
A) 3 × 3 B) 2 × 4
C) 4 × 2 D) 'yA'9("% 'Ù5 '9" "™H / Multiplication not possible
- Q.78.** x-•Y "Ù7 'Y@ 'i?'YM-'YK'ÉM'ù>•ù (Direction Cosines of x-axis) 'YM'ù> "™H•#ð
[English]: English: What are the direction cosines of the x-axis?
A) (1, 0, 0) B) (0, 1, 0)
C) (0, 0, 1) D) (1, 1, 1)
- Q.79.** '12', b±, ' Ò Ç±Â , "ù 'iA x = 0 '©0 'Y? , * "Ù0'Y> " "â 5"Ù/"Y9"é0 '©M" & "M'i?B " \$ "â 9" fð
[English]: English: How does the function f(x) = |x| behave at the point x = 0?
A) "%\$ 'B 9", * "IA •Y5'Y2'%"@ "ò ("™@•" ò 6öçF—auous but not Differentiable > •Y5'Y2'%"@ "ò & "1(" "™H / Both Continuous and Differentiable
C) ', \$ "2 8'IS "™H ' , 9" "Y "(" // Neither Continuous nor Differentiable > •Y5'Y2'%"@ "ò & "1(" "™H / Discontinuous
- Q.80.** "+ •^(" 3) (1 / (1 + x²)) dx 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: English: What is the value of "+ •^(" 3) (1 / (1 + x²)) dx?
A) < ó B) < ó`
C) < ó@ D) 2< ó0
- Q.81.** "YM'ù5" K'yK•" , ^' & B Â , & R Â ' & R "r •) \$" M'y\$ Z = 3x + 4y 'Y> •Y""ù 'l. 'é>', "Ù/"â 9"1 "ãð
[English]: English: What is the maximum value of Z = 3x + 4y subject to x + y "d 4, x "e 0, y "e 0?
A) 12 B) 16
C) 7 D) 0
- Q.82.** •ù "%?"YM'YG 'YK 3 'É> " '1>")> 'É>'I> "™H—B " 2 "i@ "M'r „tV G2' '9G 'Y@ '©M" >ù?'Y\$ "â "Ù/"â 9"1 "ð
[English]: English: A coin is tossed 3 times. What is the probability of getting exactly 2 heads?
A) 3/8 B) 1/2
C) 1/4 D) 1/8
- Q.83.** '©M" >'Y. "ù 'Y "Ù7 "â . "y " '%> "Y>'i@ ('Y "%M'ùM" "Ù "ù5"ù8"Ù) "i?'YM"y# "%?"iM'y>•) \$"1 'YG •Y(" 8"é0 "%, %G •™* 'ùA'YI
[English]: In a primary classroom setting, which pedagogical approach aligns with constructivist principles of learning?
A) '©0"ù—"é7"é•" "2 0"ù("â ò assive memorization of definitions B) "% "Ù0"ù/ •Y("Ù5"y7'2 " '%A'Ù5"é\$"Ù. 'R 8'è8"Ù/"â 8'è>'y>', ò 7F—`e expl
C) 'YG"Y2 "i?'YM"y -'YG•)&"Ù0"ù\$ "YM'ù>'iM'ù>', ò Teacher-centred & Rote learning only D) '©0"ù—"é7"é•" "2 0"ù("â ò Teacher-centred & Rote learning only
- Q.84.** 'iK "% 'iM'ù>9 'Y> 'é"%"f (HCF) 15 'l%"â 2—i8—b ,Ä4Ò 3 9"%"d 'ù&"ò 'R 8•) "Ù/"â c 9", Â \$ "2 & ")8" @ "% 'iM'ù> 'YM'ù> "™K'
[English]: HCF of two numbers is 15, LCM is 300. If one number is 60, find the other:
A) 75 B) 80
C) 90 D) 100
- Q.85.** 140 'YK •Y—"é "Ù/ 'yA'9('i' K•" "r " #'%+"" "r 0")* 'éG•" "%8"r 2"ù "â "é 'y>?
[English]: Prime factorisation of 140 is:
A) 2 × 5² × 7 B) 2² × 5 × 7
C) 2³ × 5 × 7 D) 2 × 5 × 7²
- Q.86.** '%?'éM', . "y "%G 'YL', 8" 8•) "Ù/"â 'R '©0"ù. "y/ "% 'iM'ù> (Irrational Number) "™H?
[English]: Which of the following is an irrational number?
A) " 16 B) " 9
C) " 5 D) 22/7
- Q.87.** '©0"ù. "y/ "% 'iM'ù> 17 / 8 'Y> 'i6"é2"R * "Ù0"%"> " "%8"â 9"1 "ãð
[English]: Decimal expansion of 17/8 is:
A) •Y("ù0"Ù""é0"ù\$ / Undetermined B) •Y6"é 'B "Y0"Ù\$" „æöâ×FW minating repeating)
C) •Y6"é 'B "9"%">"Y0"Ù\$" "% '©0"ù. "y/ / Non-terminating Non-repeating (Irrational)
D) •Y6"é 'B "Y0"Ù\$" „æöâ×FW minating repeating)

Q.88. 'iM"Y?"%>'B , "TM A'©& ax² + bx + c 'YG "iB'%M'ù>•) "1 'Y> 'ùK'y+"" f ± + ;"" "Ù/"â 9"1\$ "â 9" f ð

[English]: Sum of zeroes of quadratic polynomial ax² + bx + c is:

- A) -b / a B) c / a
C) b / a D) -c / a

Q.89. •ù 'iM"Y?"%>'B , "TM A'©& 'ÉM'É>'B " "ù 'É?"% "r 6")("Ù/"é 'YK•" "â /"1 4 'l%"â " #'%+"" 9" f

[English]: Find a quadratic polynomial whose sum and product of zeroes are 4 and 1:

- A) x² + 4x + 1 B) x² - 4x + 1
C) x² - 4x - 1 D) x² + 4x - 1

Q.90. "%." " # "%?'Y>'ð , 2 " ' 2 2 Ò " ' , 2 " " 2 2 " Ò "â 'R 'iM"Y?'l@'ð 9"" ... Væ— VR 6öÇWF—öâ' 9"1("r " 6" M'B "Ù/"â 9" f

[English]: Condition for unique solution of a pair of linear equations:

- A) a •/a , = b •/b , " c •/c , B) a •/a , = b •/b , = c •/c ,
C) a •/a , " b •/b , D) a •/a , = -b •/b ,

Q.91. 'É, a •/a , = b •/b , = c •/c , "TM K, 'IK "%." " # "%?'Y>'ð "r "ù\$%G "TM 2 "TM K'IG "TM H•#ð

[English]: When a •/a , = b •/b , = c •/c ,, the number of solutions is:

- A) 'iK "TM 2 / Two Distinct Solutions B) 'YK•, 9"" ("TM @•" ð æð 6öÇWF—öâ ... arallel Lines)
C) 'YG"Y2 •ù •Y&"Ü5"ù\$ / "TM 2 / Exactly One Unique Solution D) •Y(•)\$ / •Y("y "TM 2 (Infinitely many)

Q.92. k 'YG 'Y?" , "é('YG ")?•ð 8'É@'Y0'2 ' , 2 7' Ò r \$'Y> 4x + ky = 12 'Y> 'YK•, 9"" ("TM @•" 9"1 "ãð

[English]: For what value of k, the system 2x + 3y = 7 and 4x + ky = 12 has no solution?

- A) 6 B) 3
C) 2 D) 4

Q.93. 'iM"Y?"%>'B 8'É@'Y0'2 <" 2 ' , 2 2 Ò "â 5"ù5"ù "Ü\$'Y0 (D) 'YM'ù> "TM K'l> "TM H?

[English]: Discriminant of quadratic equation ax² + bx + c = 0 is:

- A) b² + 4ac B) b² - 4ac
C) 4ac - b² D) " (b² - 4ac)

Q.94. "%." " # 2x² - 4x + 3 = 0 'YG 'ÉB")K•" " *""Ü0'YC'l? 'YM'ù> "TM K'y@?

[English]: Nature of roots of 2x² - 4x + 3 = 0:

- A) "Y>"%M'l5"ù •ù5•" -"ù("Ü('ÉB"" „B â ' ò &V Â æB F—7F—æ7B)8Y>C0M'l5"ù •ù5•" 8'É>' , ")2 (D = 0) / Real and Equal Roots
C) 'YK•, 5"é8"Ü\$"Y?"R .")2 '9"9" (D < 0) D) •Y*" ?'ÉG'ò ò —' ational

Q.95. 'ù&"ð 8'É@'Y0'2 <' " Ò , 2 " Ò "r .")2 "%."é((Equal) "TM K•"Â \$ "2 2 "â "é('YM'ù> "TM K'y>?

[English]: If roots of 2x² - kx + 2 = 0 are equal, find k:

- A) 0 B) ±2
C) ±8 D) ±4

Q.96. A.P. 3, 8, 13, 18, ... 'Y> 10"Y>• *b "Ù/"â 9"1 "ãð

[English]: Find 10th term of A.P. 3, 8, 13, 18, ...:

- A) 48 B) 50
C) 45 D) 53

Q.97. '©M" %'â " , *)0"Ü# "é 'YK•" f 2 " 2 2 âââ + 100) 'Y> 'ùK'y+"" "Ù/"â 9"1 "ãð

[English]: Sum of first 100 positive integers:

- A) 5000 B) 5050
C) 5150 D) 5500

Q.98. A.P. 7, 13, 19, ..., 205 'ÉG•" " 2 'Y?'l("r *b 9"% ?

[English]: How many terms are in A.P. 7, 13, 19, ..., 205?

- A) 35 B) 33
C) 34 D) 32

Q.99. %³ABC 'ÉG•" DR "R \$2 9"%d 'ù&"ð B Ò ãR 6ÒÂ D" Ò 2 6ÒÂ R Ò 6Ò 9"2Â \$"2 T2 "â "é('YM'ù> "TM K'y>?

[English]: In %³ABC, DE "% BC. If AD=1.5, DB=3, AE=1, find EC:

- A) 3 cm B) 1.5 cm
C) 2.5 cm D) 2 cm

Q.100. 'iK "%." B'¢ \$ "Ü0"ù-" "1 'Y@ 'ÙA'É>'9 'Y> •Y(" "é\$ 4 : 9 "TM H—B "% "r "Ü7"y\$ "Ü0'12"1 'Y> •Y(" "é\$ 'YM'ù> "TM K'y>?

[English]: Sides of two similar triangles are in ratio 4:9. Ratio of their areas is:

- A) 16 : 81 B) 2 : 3
C) 4 : 9 D) 81 : 16

- Q.101.** •ù “‰.‘YK’2 \$”Ù0“ù-” ‘Y> ‘Y0”Ù# 13 “‰G’è@ ‘I0 •ù ‘ÙA’É> 5 “‰G’è@ “™H–B \$” 8“ @ ‘ÙA’É> ‘Y@ “).”Ù,“é ‘YM’ù> “™K’y@?
[English]: Hypotenuse of right triangle is 13 cm, one side is 5 cm. Other side is:
A) 10 cm B) 12 cm
C) 11 cm D) 8 cm
- Q.102.** ‘É?’‰M’iA’9 (2, 3) ‘I0 (4, 1) ‘YG ‘É@‘¢ “ &”)0” ”Ù/“â 9”1 “ ð
[English]: Distance between points (2, 3) and (4, 1) is:
A) 4 ‘é>’IM“ / 4 units B) “ 8 ‘é>’IM“ / “ 8 units
C) 2” 2 ‘é>’IM“ / 2” 2 units D) 2 ‘é>’IM“ / 2 units
- Q.103.** ‘É?’‰M’iA (-6, 8) ‘Y@ ‘éB“),“ù(”Ù&” „÷ igin) “‰G ‘iB“ @ ‘YM’ù> “™K’y@?
[English]: Distance of point (-6, 8) from origin is:
A) 6 ‘é>’IM“ / 6 units B) 14 ‘é>’IM“ / 14 units
C) 8 ‘é>’IM“ / 8 units D) 10 ‘é>’IM“ / 10 units
- Q.104.** ‘É?’‰M’iA’9 (-5, 7) ‘I0 (-1, 3) ‘YK ‘é?“>”‰G “Y>”)G “ G’i>’i “ ”r .’yM’òÙ,“ù(”Ù&” ”r (“ù0”Ù&”y6“é ‘R ”Ù/“â 9”1 ‘yG?
[English]: Mid-point of line segment joining (-5, 7) and (-1, 3):
A) (-3, 5) B) (-2, 5)
C) (3, -5) D) (-3, -5)
- Q.105.** $\sin 60^\circ \times \cos 30^\circ + \cos 60^\circ \times \sin 30^\circ$ ‘Y> ‘é>’, ”Ù/“â 9”1 “ãð
[English]: Value of $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$ is:
A) 1/2 B) 1
C) “ 3/2 D) 0
- Q.106.** $9 \sec^2 A - 9 \tan^2 A$ ‘Y> ‘é>’, “ù8’YG ‘É0“é,“ 9”fð
[English]: $9 \sec^2 A - 9 \tan^2 A$ equals:
A) 8 B) 1
C) 9 D) 0
- Q.107.** $\tan 65^\circ / \cot 25^\circ$ ‘Y> ‘é>’, ”Ù/“â 9”1 “ãð
[English]: Value of $\tan 65^\circ / \cot 25^\circ$ is:
A) 1/” 3 B) 0
C) “ 3 D) 1
- Q.108.** •ù •©0”Ù”Ù5“é” •)-”r “ “é •™8‘Y@ •© ‘©>•, ”r “ >’É0 “™H–B 8”)0”Ù/ ‘Y> •™(”Ù(“ù(‘YK’2 „ævÆR öb VÆPvation) ‘YM’ù> “™K’y@?
[English]: Shadow of vertical pole equals its height. Angle of elevation is:
A) 45° B) 30°
C) 60° D) 90°
- Q.109.** ‘ÙB’é? ‘YG •ù ‘É?•)&” 8”r .” (“é0 ‘YG ‘©>’b 8”r 3 .” “ &”)0” “ “ (“é0 ‘YG “i?’i0 ‘Y> •™(”Ù(“ù(‘YK’2 3 9”‰d ‘é@’‰>“ ” •)
[English]: Angle of elevation is 30° from 30m away. Height of tower is:
A) 30” 3 m B) 10” 3 m
C) 15 m D) 20 m
- Q.110.** “YC’IM’B ”r “ù8” ,“é9”Ù/ ‘É?•)&” 8”r “ ‘©@ ‘y ‘iK “‰M’©0”Ù6 “ G’i>’9 PA ‘I0 PB ‘Y@ “).”Ù,“é ‘ùK•” .”y ‘YM’ù> “‰ ‘É ‘r 9”1
[English]: Relation between lengths of two tangents drawn from external point:
A) PA < PB B) PA > PB
C) PA = PB (“‰. “é(“™K’I@ “™H•”• D) PA = 2PB
- Q.111.** •Y0”Ù”YC’IM’B .”y ‘É(“â ”1# (Angle in a semi-circle) ‘Y?’I(“â 9”1\$“â 9”fð
[English]: Angle in a semicircle is:
A) 45° B) 60°
C) 180° D) “‰.‘YK’2 f“ ’ ò &–v‡B ævÆR f“ •
- Q.112.** ‘ù&“ò 8”Ù” M“b 0”y “â A ‘I0 PB •ù ‘iB“‰0”r 8”r f ”r ”1# ‘©0 ‘ÙA’Y@ “™K•”Â \$”2 “ ðA ‘Y> ‘é>’, ”Ù/“â 9”1 “â „ò ”y ‘iM“ 9”“ð
[English]: If tangents PA and PB are inclined at 80°, find “ POA:
A) 50° B) 60°
C) 70° D) 80°
- Q.113.** ‘IM“ ?‘ÉM’ù> r “Y>”)G “YC’IM’B ”r “ , \$”Ù0“ù ”Ù/i “ â “Ù7”y\$”Ù0”2 ‘YM’ù> “™K’y> ‘É?’‰ “â ”1# „, 9”fð
[English]: Area of a sector of angle ;, öb 6—&6ÆR öb adius r:
A) (,; ò f ‘r Àr² B) (,; ò 3c ‘r Àr²
C) (,; ò 3c ‘r #Àr D) 2< +

- Q.114.** •ù “YC’IM'B ” * “ ?y? ‘IO ‘YM“yG’IM“ + “ 8•) ”Ù/“é\$”Ù.‘R 0”) * “%G ‘É0“é,“ 9”% –B “, 5”9\$”Ù\$ ‘Y@ ‘IM“ ?‘ÉM’ù> ‘YM’ù> “™K
 [English]: If circumference and area of a circle are numerically equal, radius is:
 A) 4 ‘é>’IM“ / 4 units B) < .“é\$”Ù0‘R ò À units
 C) 2 ‘é>’IM“ / 2 units D) 7 ‘é>’IM“ / 7 units
- Q.115.** ‘IM“ ?‘ÉM’ù> r “Y>”)G •ù ‘K, “ M’y ”12”r “â 8‘ém’©B“ M’2 *”97”Ù ” / ‘YM“yG’IM“ + “ ”Ù/“â 9”1\$“â 9”f ð
 [English]: Total surface area of a solid hemisphere of radius r:
 A) $2/3 < +0$ B) $2 < +$
 C) $4 < +$ D) $3 < +$
- Q.116.** ‘ù&”ò ‘R ”12”r ” \$”Ù0“ù ”Ù/“â &”1 ” (“ “ &” “é, ‘IK •™8‘Y> •i/’I(‘Y?’I(”r ” (“â 9”2 “é ‘y>?
 [English]: If radius of a sphere is doubled, its volume becomes:
 A) 8 ‘yA’%> / 8 times B) 4 ‘yA’%> / 4 times
 C) 2 ‘yA’%> / 2 times D) 16 ‘yA’%> / 16 times
- Q.117.** “%.”é(‘IM“ ?‘ÉM’ù> ‘IO “%.”é(•© ‘©>•, 5“é2”r ‘R ,y2”, “ ‘R 6•) ” ”r ‘ù\$%K•” “â ‘%A’©>’B ”Ù/“â 9”1 “ã ð
 [English]: Ratio of volumes of a cylinder and cone of same radius and height:
 A) 1 : 3 B) 3 : 1
 C) 1 : 1 D) 2 : 1
- Q.118.** ‘é>yM’ðÂ .“é”Ù/’R “ , “™A”) ‘YG ‘É@‘¢ 8“™@ •i(” -“Y?’R 8•),•) (Empirical Relationship) ‘YM’ù> “™H?
 [English]: Correct empirical relation between Mean, Median and Mode:
 A) $3 ‘É9” 2‘R Ò .“é”Ù/’R Ò .“é”Ù/$ B) $‘É9” 2‘R Ò .“é”Ù/’R Ò 2 .“é”Ù/ / \text{Mode} = 2 \text{ Median} - 3 \text{ Mean}$
 C) $‘É9” 2‘R Ò 2 .“é”Ù/’R Ò .“é”Ù/ / \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$ D) $‘é>yM’ù = 3 ‘É9” 2‘R Ò .“é”Ù/$
- Q.119.** “% ‘iM’ù>9 2, 4, 6, 8, 10 ‘Y> “%.”é ‘IO ‘é>yM’ò ”Ù/“â 9”1 “ã ð
 [English]: Arithmetic mean of 2, 4, 6, 8, 10:
 A) 8 B) 5
 C) 7 D) 6
- Q.120.** •i ‘Y!’ÉK•” rÂ “Â Â BÂ ‘Â RÂ , ” .“é”Ù/“ù “â „ÖVF– â” ”Ù/“â 9”1 ” ð
 [English]: Median of data 7, 2, 11, 4, 9, 5, 8:
 A) 7 B) 8
 C) 6 D) 5
- Q.121.** ‘É ‘ù(2, 3, 5, 3, 7, 3, 8, 3, 9 ‘Y> ‘É9” 2‘R ”Ù/“â 9”1 “ã ð
 [English]: Mode of the distribution 2, 3, 5, 3, 7, 3, 8, 3, 9 is:
 A) 5 B) 3
 C) 7 D) 9
- Q.122.** ‘Y?’%@ •Y8•)-“R ‘ù(“â „–x ÷76– le Event) ‘Y@ ‘©M“ >’ù?’Y\$“â ”Ù/“â 9”1\$” 9”f ð
 [English]: Probability of an impossible event is:
 A) 0.5 B) 1
 C) 0 D) -1
- Q.123.** •ù ‘©>“%G ‘YK •ù ‘É>“ +”y ‘Y(”r * “ ‘Ù>‘ÉM’ð 8•) ”Ù/“â ... ime Number) •i(”r ” *”Ù0“é/“ù ‘I> ‘YM’ù> “™K’y@?
 [English]: Probability of getting a prime number in a single throw of a die:
 A) $2 / 3$ B) $1 / 3$
 C) $1 / 6$ D) $1 / 2$
- Q.124.** •Y “Ù ” \$“ 9 ‘1G•) ” •, S” *’IM’IK•” ” \$“é6 ‘Y@ ‘y!”Ù!” .”y “%G •ù ‘©\$”Ù\$“â (“ù “é2”â “é\$“â 9”%d •ù “)>“ 0•) ‘Y> ‘É>i6“é9
 [English]: Probability of drawing a red king from a deck of 52 cards:
 A) $1 / 26$ B) $1 / 52$
 C) $1 / 13$ D) $2 / 13$
- Q.125.** •ù ‘Y>“ 8”r ” \$’R c “ù.” ù •) “â 8”r “é\$” 9”, \$’Y> 40 ‘Y?’é@/’% “ù> “%G “)L’ù\$” 9”%d ‘©B“ @ ‘ù>’IM“ > ‘Y@ ‘I8’B “é2 ‘YM’
 [English]: Car travels at 60 km/h and returns at 40 km/h. Average speed is:
 A) $50 ‘Y?’é@/’% “ù>$ B) $48 ‘Y?’é@/’% “ù>$
 C) $45 ‘Y?’é@/’% “ù>$ D) $52 ‘Y?’é@/’% “ù>$
- Q.126.** 15,000 ‘Y@ “ >“i? ‘©0 10% “Y>“ M“y?’R &“ 8”r ” 5“ M“r ”r 4’ “ 4’ “â •) \$“ ”Ù/“â 9”1 “ã ð
 [English]: Difference between CI and SI on 15000 at 10% for 2 years:
 A) 1100 B) 125
 C) 150 D) 175

- Q.127.** A 'Y?'%@ 'Y>' M'ò "² " &"ù('éG•" \$'Y> B 18 'i?', .y '©B" > 'Y0'I> "™H-B &"¹("¹ 'é?") " "ù\$%G 'i?', .y '©B" > 'Y0'y 'yG?
[English]: A does work in 12 days, B in 18 days. Together in how many days?
A) 7.5 'i?', ò rãR F ys B) 8 'i?', ò , F ys
C) 6 'i?', ò b F ys D) 7.2 'i?', ò rã" F ys
- Q.128.** 'ù&"ò " (" "r .")²"Ù/ 'éG•" #RR " 5"9&"ù""ò 9"² "é , 'IK 'yC"™?"9@ 'YK 'i*B .y 'Y?'I("r *""Ù0'I?"i\$ 'Y@ 'Y." " (" 9"¹ " \$"é "ò
[English]: Price of sugar rises by 25%. Reduction in consumption to keep expenditure constant:
A) 20% B) 25%
C) 16.66% D) 15%
- Q.129.** 'ùB'YM")?' 5"ù-"é ' , *""Ù0'ég'ù?'Y> a = bq + r 'éG•" 6"y7"¹2 r 'YG ")?•ò "É("%@ "i0"Ù\$ "%\$""Ù/ "™H?
[English]: In Euclid's division lemma a = bq + r, condition for remainder r is:
A) 0 < r "d b B) 0 "d r < b
C) 0 "d r "d b D) r > b
- Q.130.** "%,%G '¹K'ù@ •Y-"é "Ù/ "% 'iM'ù> (Smallest Prime Number) 'YL', 8" 9"fð
[English]: Smallest prime number is:
A) 3 B) 1
C) 2 (ù 'é>'IM" 8'â 'Ù>'ÉM'ò' ò " „öæÇ' W`en Prime) D) 0
- Q.131.** " H'i?'R , "™A'©& (Linear Polynomial) 'Y@ •Y""ù 'I. '%>'B "ù\$%@ "™K'I@ "™H?
[English]: Degree of a linear polynomial is:
A) 3 B) 2
C) 0 D) 1
- Q.132.** 'ù&"ò &"Ù5"ù "é\$ 'É9" *b <" Ò 7, ² " "r 6")("Ù/"é 'R ± 'I0 , " 9"¹ , 'IK 1/; ² ó² 'Y> 'é>', "Ù/"â 9"¹ "ãð
[English]: If ; $\hat{A}^2$ are zeroes of $x^2 - 3x + 2$, find $1/;^2 \hat{\alpha}^2$:
A) 3 / 2 B) 2 / 3
C) -3 / 2 D) 1
- Q.133.** 'ù&"ò &"² 0"y "é •" 'R , "ù 'iA '©0 '©M" \$"ù "Ù "y& 'Y0'I@ "™H•"Â \$"² 5"" , "ù 'iA 'iK'%K•" 8'é@ 'Y0'9K•" "â "‰8"â 9"" 9"¹ "ãð
[English]: If two lines intersect at a point, that point is:
A) •Y(•)\$ •Y("y "™2 / Infinitely Many Solutions B) •Y&"Ù5"ù\$" / "™2 (Unique Solution)
C) 'YK•, 9"" ("™@•" ò æð 6öÇWF-öâ ... arallel Lines) D) 'iK "™2 / Two Distinct Solutions
- Q.134.** 'ù&"ò , ² ' Ò B \$'Y> x - y = 4 "™K, 'IK x 'I0 y 'YG 'é>', "Ù/"â 9"¹ 'yG?
[English]: Solve x + y = 14 and x - y = 4:
A) x = 10, y = 4 B) x = 8, y = 6
C) x = 9, y = 5 D) x = 7, y = 7
- Q.135.** 'iM"Y?'%>'B 8'é@ 'Y0'2 "â ."é('R 0")* (Standard Form) 'YM'ù> "™K'I> "™H?
[English]: Standard form of quadratic equation is:
A) $ax^2 + c = 0$ B) $ax + b = 0$
C) $ax^3 + bx^2 + cx + d = 0$ D) $ax^2 + bx + c = 0$ ('É9"é a "" 0)
- Q.136.** "%." " # $x^2 - 9 = 0$ 'YG 'éB"" "Ù/"â 9"¹ 'yG?
[English]: Roots of equation $x^2 - 9 = 0$ are:
A) ±3 B) 9
C) 3 D) ±9
- Q.137.** A.P. -10, -6, -2, 2, ... 'Y> "%>" M"R •)\$" †B' "Ù/"â 9"fð
[English]: Common difference of A.P. -10, -6, -2, 2, ... is:
A) -4 B) 4
C) 2 D) -2
- Q.138.** '©M" %'â â 5"ù7'â *""Ù0"é "9\$ "% 'iM'ù>9 'Y> 'ùK'y+"" ...7VÒ öb f—'7B â öFB æ GW al numbers) 'YM'ù> "™K'I> "™H?
[English]: Sum of first n odd natural numbers is:
A) 2n B) n(n+1)
C) n² D) n(n+1)/2
- Q.139.** 'É?'%M'iA (-4, -5) 'Y?' , 'IA" M'Y>•)6 'éG•" 8"Ù% "ù\$ "™H?
[English]: In which quadrant does point (-4, -5) lie?
A) '©\$ 0"Ù% '©\$ 0"Ù%é "b ò `ourth Quadrant (+,-) B) '©M" %'â 'IA" M'Y>•)6 / First Quadrant (+,+)
C) 'iM"Y?'I@'ò 'IA" M'Y>•)6 / Second Quadrant (-,+)
D) 'IC'I@'ò 'IA" M'Y>•)6 (Third Quadrant)

- Q.140.** 'ù&"ð \$" ('É?'%M'IA "% " G'i@ (Collinear) "™K•"Â \$"² '%8"r , '%("r 5"é2"r \$"Ù0"ù-" 'Y> 'YM"yG'IM" +"" "Ù/"â 9"¹ "ãð
[English]: Area of triangle formed by three collinear points is:
A) 0 B) 1
C) -1 D) •Y* " ?'é?'B 0")* "%G •Y("y / Infinitely Many
- Q.141.** 'IM" ?'ÙA'Â "r 6" 0"Ù7 (x •, y •), (x ,, y ,), (x f, y f) "™K•"Â \$"² "y("Ù&"Ù0'R "r ("ù0"Ù&"y6"é 'R "Ù/"â 9"¹ 'yG?
[English]: Coordinates of centroid of triangle with given vertices:
A) ((x •+x ,)/2, (y •+y ,)/2) B) ((x •+x ,+x f)/3, (y •+y ,+y f)/3)
C) (x •+x ,+x f, y •+y ,+y f) D) (x •x ,x f, y •y ,y f)
- Q.142.** 'ù&"ð 6-â , = 3/5 "™K, 'IK cos ;, "â ."é('YM'ù> "™K'y>?
[English]: If sin ;, Ò 2óRÂ f-æB 6÷2 ,:
A) 3 / 4 B) 5 / 4
C) 4 / 5 D) 4 / 3
- Q.143.** tan 45° 'Y> 'é>', "Ù/"â 9"¹\$ "â 9" fð
[English]: Value of tan 45° is:
A) " 3 B) 0
C) 1/" 3 D) 1
- Q.144.** 1 + cot² ;, "ù8'YG 'É0"é," 9"¹\$ "â 9" fð
[English]: 1 + cot² ;, —2 W V Â Fó
A) cosec² ;€ B) sec² ;€
C) tan² ;€ D) sin² ;€
- Q.145.** "YC'IM'B "r 'R ,"ù 'iA '©0 'Y?'I(" 8"Ù* M"b 0"y "é •" " '©@ 'É> "% 'l@ "™H•#ð
[English]: How many tangents can be drawn at a point on a circle?
A) 2 B) 'YG"Y2 1 (Only one)
C) •Y(•)\$ / Infinity (") D) 0
- Q.146.** •ù "YC'IM'B " "ù\$'%@ "%."é 'l0 "%M'©0"Ù6 " G'i>•ù (Parallel Tangents) "™K "% 'l@ "™H•#ð
[English]: Maximum number of parallel tangents a circle can have:
A) 1 B) 4
C) 2 D) •Y(•)\$ / Infinity (")
- Q.147.** 'IM" ?'ÉM'ù> r "Y>")G "YC'IM'B "r 'R 'lA" M'Y>•)6 (Quadrant) 'Y> 'YM"yG'IM" +"" "Ù/"â 9"¹\$ "â 9" fð
[English]: Area of a quadrant of a circle of radius r is:
A) 2< B) 1/2 < +
C) < + D) 1/4 < +
- Q.148.** 'IM" ?'ÉM'ù>9 R 'l0 r (R > r) "Y>")G 'iK "% 'YG•&)"Ù0" / "YC'IM'IK•" "r , " 'É("r 5")/ (Ring) 'Y> 'YM"yG'IM" +"" "Ù/"â 9"¹\$ "â 9" fð
[English]: Area of ring between concentric circles of radii R and r:
A) < ...+" Ò + "• B) < ..." Ò ">
C) 2< ..." Ò "• D) < + " ò +
- Q.149.** 'IM" ?'ÉM'ù> 7 "%G'é@ 'l% "â •) "é 10 "%G'é@ "Y>")G ")."Ù,"YC'IM'l@'ò , "y2', "â 'ù\$, "Ù/"â 9"¹ "ãð
[English]: Volume of cylinder with r = 7 cm, h = 10 cm:
A) 1450 "%G'é@³ / 1450 cm³ B) 1540 "%G'é@³ / 1540 cm³
C) 1500 "%G'é@³ / 1500 cm³ D) 770 "%G'é@³ / 770 cm³
- Q.150.** "i 'YA 'Y@ 'l?' M'ù •© '©>•, ...6Æ çB †V-v‡B vÂr' "â 8")\$ "Ù0 'YM'ù> "™H?
[English]: Slant height of cone with radius r and height h:
A) " (r² - h²) B) r² + h²
C) " (r² + h²) D) r + h
- Q.151.** 'ÙA'É> a "Y>")G •ù '%((Cube) 'YG 'éA'iM'ò 5"ù " M'2 „F- vöæ Â' " 2'ém'É>•, "Ù/"â 9"¹ " ð
[English]: Length of diagonal of a cube of edge a is:
A) a / " 3 B) a" 2
C) 3a D) a" 3
- Q.152.** "Y0"Ù •Y 'l0"é2 10 - 25 'Y> "Y0"Ù '©?"™M, „6Æ 72 Ö k) 'YM'ù> "™K'y>?
[English]: Class mark of class interval 10 - 25 is:
A) 17.5 B) 15
C) 20 D) 35

Q.153. 'i 'Y!'ÉK•" "Â #RÂ RÂ ,Â 3RÂ # "â * " ?"%0 (Range) 'YM'ù> "TMK'y>?

[English]: Range of data 12, 25, 15, 18, 35, 20 is:

- A) 35 B) 23
C) 12 D) 20

Q.154. 'Y?'%0@ '%0?'iM'©?'B 'ù("â ...7W&R W`ent) 'Y@ '©M" >'ù?'Y\$â "ù\$'%0@ "TMK'I@ "TMH?

[English]: Probability of a sure event is:

- A) 0.5 B) 0
C) 1 D) •Y("ù6"Ù "ù\$ / Indeterminate

Q.155. 'ù&"ò „R' Ò ã R 9"2Â \$"2 tR ("TM@•"r ... †æ÷B R" " *"Ù0"é/"ù 'l> 'YM'ù> "TMK'y@?

[English]: If $P(E) = 0.05$, find $P(\text{not } E)$:

- A) 1.05 B) 0.05
C) 0.50 D) 0.95

Q.156. 'iK "%0?'YM'YK•" "2 'R 8"é% •TM "é2'%0G '©0 'Y. "%0G 'Y. •ù '©?'iM'B „ B ÆV 7B öæR †V B' '%0G 'Y@ '©M" >'ù?'Y\$â "Ù/"â

[English]: Probability of getting at least one head when tossing two coins:

- A) 3 / 4 B) 1 / 4
C) 1 / 2 D) 1

Q.157. •ù "Y8"ù\$" "2 "s# .y 'ÉG'©("r "â # R "â 2"é- "TMK'l> "TMH-B 5"%0M'IA 'Y> 'YM" / 'éB")M'ò „6÷7B ice) 'YM'ù> "TMK'y>?

[English]: Selling at 1720 gives 20% profit. Cost price is:

- A) 1580 B) 1600
C) 1640 D) 1560

Q.158. 120 'é@'ù0 " 'É@ 'ùM" G', SB "ù." ù •) "â " "é2 "%0G •ù 'i 'ÙG 'YK 'Y?'l("r 8'é/ 'éG•" *"é0 'Y0"y " ð

[English]: 120m long train running at 54 km/h crosses a pole in:

- A) 12 "%0G'Y ' ò " 6V6öæG0 B) 6 "%0G'Y ' ò b 6V6öæG0
C) 10 "%0G'Y ' ò 6V6öæG0 D) 8 "%0G'Y ' ò , 6V6öæG0

Q.159. (7 y u - 3 u x) 'Y> •y "é 'Y> •Y 'R "Ù/"â 9"1 "ãð

[English]: Unit digit of (7 y u - 3 u x) is:

- A) 4 B) 6
C) 0 D) 7

Q.160. 'Y?'l("r 8'é/ 'éG•" "1 'y(5% "Y>" M"y?'R 8"é""é0'2 , "Ù/"é 'Y@ 'i0 "%0G 'iK'yA'%0> "TMK 'É>•ù "ãð

[English]: In what time will a sum double at 5% simple interest per annum?

- A) 10 "Y0"Ù7 / 10 Years B) 20 "Y0"Ù7 / 20 Years
C) 15 "Y0"Ù7 / 15 Years D) 25 "Y0"Ù7 / 25 Years

Q.161. •ù "%0.'É>"TMA 'IM" ?'ÙA'Â " *"Ù0'IM'ùG'R -" "â b 8"y." 9"%0d •y8'Y> 'YM"yG'IM" +"" "Ù/"â 9"1 "ãð

[English]: Area of an equilateral triangle with side 6 cm is:

- A) 36 "%0G'é@2 / 36 cm² B) 18" 3 "%0G'é@2 / 18" 3 cm²
C) 9" 3 "%0G'é@2 / 9" 3 cm² D) 6" 3 "%0G'é@2 / 6" 3 cm²

Q.162. 3/4, 5/6, 7/8, 9/10 'éG•" 8'É8"r , ' <" -"ù("Ù('YL', 8" 9" fð

[English]: Largest fraction among 3/4, 5/6, 7/8, 9/10:

- A) 3 / 4 B) 7 / 8
C) 5 / 6 D) 9 / 10

Q.163. '©M" %'â S 8'â *"Ù0"é "9\$ "%0 'iM'ù>'9 'Y> 'l8'B "Ù/"â 9"1 "ãð

[English]: Average of first 50 even natural numbers is:

- A) 51 B) 50
C) 50.5 D) 100

Q.164. 'ù&"ò %â†, Ò ' 2 %â†, 2 ' Ò 3# 9"2Â \$"2 , "â ."é('YM'ù> "TMK'y>?

[English]: If $2^x(x-1) + 2^x(x+1) = 320$, find x:

- A) 6 B) 7
C) 8 D) 5

Q.165. 20% 'l0 10% 'Y@ 'iK 'YM" ."ù '1B'ò ...7V66W76—`e Discounts) 'YG "%0.'IA")M'ò 'Y2 '1B'ò "Ù/"â 9"1 " ð

[English]: Single equivalent discount for successive discounts of 20% and 10%:

- A) 25% B) 30%
C) 28% D) 22%

- Q.166.** "i>•)\$ 'É2 'éG•" ("é5 'Y@ '©>"" "ù." ù •) "â \$'Y> 'y>" > 'Y@ '©>"" " "ù." ù •) "â 9"%d 'y>" > 'Y@ 'i?'i> 'éG•" "é2 'YM'ù> "TMK
[English]: Speed in still water 10 km/h, stream 2 km/h. Downstream speed is:
A) 14 'Y?'é@/'% "ù> B) 8 'Y?'é@/'% "ù>
C) 10 'Y?'é@/'% "ù> D) 12 'Y?'é@/'% "ù>
- Q.167.** '©>•y* A 'ù 'Y@ 'YK 10 '% 'ùG 'l%"â *é '¢ " R •) "r.'y 'Ù0'I> "TMH-B &"1("1 'é?") " "ù\$'%G "%."ò."y 'Ù0'y 'yG?
[English]: Pipe A fills tank in 10h, Pipe B in 15h. Together in:
A) 6 '% 'ùG / 6 Hours B) 8 '% 'ùG / 8 Hours
C) 5 '% 'ùG / 5 Hours D) 7.5 '% 'ùG / 7.5 Hours
- Q.168.** 'ÉH•) "ù 'r * " @ 'YM"y> 'YG "% 'i0"Ù- 'éG•" !"y "â •) " * "Ù0"ù "y6", " 8•) "Ù/"és"Ù.'R '9("â * " 'y>" ?'B * "Ù0"iM', "â 8"TM @ •TM\$
[English]: Solve the quantitative/logical/linguistic problem: In a banking assessment scenario concerning Data Interpretation & Quadratic Equations, select the valid resolution.
A) "%M'Y?'I? Alpha 'é>'%M'ò 9", ò 6öæF—F—ôâ Ç † †öÆG2 Vâ) "%M'Y?'I? Beta "% " 'I0 '©M" >é>'9?'R 9"" 9", ò 6öæF—F—ôâ &WF —2 F
C) 'iK"%K•" 8"Ù0"ù\$ "ù/"é •Y* " M'ù>'©M'B 9"% / Both Alpha and Beta are insufficient / Data is insufficient
- Q.169.** Which of the following statements concerning Relationship between Zeroes & Coefficients is correct?
A) Statement regarding Relationship between Zeroes & Coefficients is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.170.** Which of the following statements concerning Consistency of Equations is correct?
A) Statement regarding Consistency of Equations is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.171.** Which of the following statements concerning Factorization Method is correct?
A) Statement regarding Factorization Method is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.172.** Which of the following statements concerning Sum of First n Terms is correct?
A) Statement regarding Sum of First n Terms is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.173.** Which of the following statements concerning Distance Formula is correct?
A) Statement regarding Distance Formula is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.174.** Which of the following statements concerning Complementary Angle Ratios is correct?
A) Statement regarding Complementary Angle Ratios is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.175.** Which of the following statements concerning Concentric Circles is correct?
A) Statement regarding Concentric Circles is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.176.** Which of the following statements concerning Mean, Median and Mode of Grouped Data is correct?
A) Statement regarding Mean, Median and Mode of Grouped Data is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.177.** Which of the following statements concerning Division Algorithm is correct?
A) Statement regarding Division Algorithm is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.178.** Which of the following statements concerning Graphical Method of Solution is correct?
A) Statement regarding Graphical Method of Solution is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.179.** Which of the following statements concerning Completing the Square is correct?
A) Statement regarding Completing the Square is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above
- Q.180.** Which of the following statements concerning Word Problem Applications is correct?
A) Statement regarding Word Problem Applications is canonically valid and the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid D) None of the above

Q.181. Which of the following statements concerning Section Formula & Midpoint is correct?

- A) Statement regarding Section Formula & Midpoint is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.182. Which of the following statements concerning Angle of Elevation and Depression is correct?

- A) Statement regarding Angle of Elevation and Depression is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.183. Which of the following statements concerning Tangent at any Point of Circle is correct?

- A) Statement regarding Tangent at any Point of Circle is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.184. Which of the following statements concerning Empirical Probability is correct?

- A) Statement regarding Empirical Probability is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.185. Which of the following statements concerning Geometrical Meaning of Zeroes is correct?

- A) Statement regarding Geometrical Meaning of Zeroes is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.186. Which of the following statements concerning Substitution & Elimination is correct?

- A) Statement regarding Substitution & Elimination is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.187. Which of the following statements concerning Quadratic Formula & Nature of Roots is correct?

- A) Statement regarding Quadratic Formula & Nature of Roots is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.188. Which of the following statements concerning General nth Term of an AP is correct?

- A) Statement regarding General nth Term of an AP is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.189. Which of the following statements concerning Area of a Triangle is correct?

- A) Statement regarding Area of a Triangle is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.190. Which of the following statements concerning Trigonometric Ratios & Identities is correct?

- A) Statement regarding Trigonometric Ratios & Identities is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.191. Which of the following statements concerning Length of Tangents from External Point is correct?

- A) Statement regarding Length of Tangents from External Point is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.192. Which of the following statements concerning Complementary Events is correct?

- A) Statement regarding Complementary Events is canonically valid and the secondary observation holds true under standard syllabus specifications
B) Only the secondary observation holds true under standard syllabus specifications
C) Both statements are factually invalid
D) None of the above

Q.193. Relationship between Zeroes & Coefficients 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd

[English]: Which of the following statements concerning Relationship between Zeroes & Coefficients is correct?

- A) Relationship between Zeroes & Coefficients 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 9'IM'ò 9"fd / Both statements are canonically valid and correct
B) Only the secondary observation holds true under standard syllabus specifications
C) 'iK'%K" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"% / Both statements are factually invalid
D) 'iK'%K" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"% / Both statements are factually invalid

Q.194. Consistency of Equations 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd

[English]: Which of the following statements concerning Consistency of Equations is correct?

- A) Consistency of Equations 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 9'IM'ò 9"fd / Both statements are canonically valid and correct
B) Only the secondary observation holds true under standard syllabus specifications
C) 'iK'%K" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"% / Both statements are factually invalid
D) 'iK'%K" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"% / Both statements are factually invalid

Q.195. Factorization Method 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd

[English]: Which of the following statements concerning Factorization Method is correct?

- A) Factorization Method 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 9'IM'ò 9"fd / Both statements are canonically valid and correct
B) Only the secondary observation holds true under standard syllabus specifications
C) 'iK'%K" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"% / Both statements are factually invalid
D) 'iK'%K" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"% / Both statements are factually invalid

- Q.196.** Sum of First n Terms 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Sum of First n Terms is correct?
A) Sum of First n Terms 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.197.** Distance Formula 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Distance Formula is correct?
A) Distance Formula 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.198.** Complementary Angle Ratios 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Complementary Angle Ratios is correct?
A) Complementary Angle Ratios 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.199.** Concentric Circles 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Concentric Circles is correct?
A) Concentric Circles 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.200.** Mean, Median and Mode of Grouped Data 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Mean, Median and Mode of Grouped Data is correct?
A) Mean, Median and Mode of Grouped Data 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.201.** Division Algorithm 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Division Algorithm is correct?
A) Division Algorithm 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.202.** Graphical Method of Solution 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Graphical Method of Solution is correct?
A) Graphical Method of Solution 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.203.** Completing the Square 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Completing the Square is correct?
A) Completing the Square 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.204.** Word Problem Applications 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Word Problem Applications is correct?
A) Word Problem Applications 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.205.** Section Formula & Midpoint 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Section Formula & Midpoint is correct?
A) Section Formula & Midpoint 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.206.** Angle of Elevation and Depression 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Angle of Elevation and Depression is correct?
A) Angle of Elevation and Depression 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.207.** Tangent at any Point of Circle 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Tangent at any Point of Circle is correct?
A) Tangent at any Point of Circle 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications
- Q.208.** Empirical Probability 'YG "% i0"Ù- 'ég•' ("ù."Ù(")?i?'B .y "%G 'YL', 8"â 5"ù ")M'ç 8'IM'ò 9" fđ
[English]: Which of the following statements concerning Empirical Probability is correct?
A) Empirical Probability 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$•2 8M'ç 8'IM'ò 9" fđ / Both statements are invalid
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are invalid
standard syllabus specifications

- Q.209. Geometrical Meaning of Zeroes 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Geometrical Meaning of Zeroes is correct?
A) Geometrical Meaning of Zeroes 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Geometrical Meaning of Zeroes 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.210. Substitution & Elimination 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Substitution & Elimination is correct?
A) Substitution & Elimination 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Substitution & Elimination 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.211. Quadratic Formula & Nature of Roots 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Quadratic Formula & Nature of Roots is correct?
A) Quadratic Formula & Nature of Roots 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Quadratic Formula & Nature of Roots 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.212. General nth Term of an AP 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning General nth Term of an AP is correct?
A) General nth Term of an AP 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) General nth Term of an AP 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.213. Area of a Triangle 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Area of a Triangle is correct?
A) Area of a Triangle 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Area of a Triangle 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.214. Trigonometric Ratios & Identities 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Trigonometric Ratios & Identities is correct?
A) Trigonometric Ratios & Identities 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Trigonometric Ratios & Identities 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.215. Length of Tangents from External Point 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Length of Tangents from External Point is correct?
A) Length of Tangents from External Point 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Length of Tangents from External Point 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.216. Complementary Events 'YG "% i0"Ù- 'éG•' ("ù."Ù(")?i?B .y "%G 'YL', 8"â 5"ù ")M'¢ 8'IM'ò 9"fd**
[English]: Which of the following statements concerning Complementary Events is correct?
A) Complementary Events 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
B) Complementary Events 'YG "% 'É 'r .y '©M" %â 'Y('©B" M'9\$2 8IM'ò 9"fd and correct
C) 'iK'‰K•" 'Y(•Y8'IM'ò "Y 'IM" A'ù?'©B" M'2 9"‰ / Both statements are mutually invalid G 'YK•, ("™@•" ò æöæR öb F†R &ðve
- Q.217. With reference to Relationship between Zeroes & Coefficients, which of the following statements is correct?**
A) The foundational proposition regarding Relationship between Zeroes & Coefficients is empirically valid under standard board conditions
B) Only theoretical statements are contradicted by standard evidence, none of the above
C) Both theoretical statements are contradicted by standard evidence, none of the above
- Q.218. With reference to Consistency of Equations, which of the following statements is correct?**
A) The foundational proposition regarding Consistency of Equations is empirically valid under standard board conditions
B) Only theoretical statements are contradicted by standard evidence, none of the above
C) Both theoretical statements are contradicted by standard evidence, none of the above
- Q.219. With reference to Factorization Method, which of the following statements is correct?**
A) The foundational proposition regarding Factorization Method is empirically valid under standard board conditions
B) Only theoretical statements are contradicted by standard evidence, none of the above
C) Both theoretical statements are contradicted by standard evidence, none of the above
- Q.220. With reference to Sum of First n Terms, which of the following statements is correct?**
A) The foundational proposition regarding Sum of First n Terms is empirically valid under standard board conditions
B) Only theoretical statements are contradicted by standard evidence, none of the above
C) Both theoretical statements are contradicted by standard evidence, none of the above
- Q.221. With reference to Distance Formula, which of the following statements is correct?**
A) The foundational proposition regarding Distance Formula is empirically valid under standard board conditions
B) Only theoretical statements are contradicted by standard evidence, none of the above
C) Both theoretical statements are contradicted by standard evidence, none of the above
- Q.222. With reference to Complementary Angle Ratios, which of the following statements is correct?**
A) The foundational proposition regarding Complementary Angle Ratios is empirically valid under standard board conditions
B) Only theoretical statements are contradicted by standard evidence, none of the above
C) Both theoretical statements are contradicted by standard evidence, none of the above

- Q.223.** With reference to Concentric Circles, which of the following statements is correct?
 A) The foundational proposition regarding Concentric Circles is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.224.** With reference to Mean, Median and Mode of Grouped Data, which of the following statements is correct?
 A) The foundational proposition regarding Mean, Median and Mode of Grouped Data is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.225.** With reference to Division Algorithm, which of the following statements is correct?
 A) The foundational proposition regarding Division Algorithm is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.226.** With reference to Graphical Method of Solution, which of the following statements is correct?
 A) The foundational proposition regarding Graphical Method of Solution is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.227.** With reference to Completing the Square, which of the following statements is correct?
 A) The foundational proposition regarding Completing the Square is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.228.** With reference to Word Problem Applications, which of the following statements is correct?
 A) The foundational proposition regarding Word Problem Applications is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.229.** With reference to Section Formula & Midpoint, which of the following statements is correct?
 A) The foundational proposition regarding Section Formula & Midpoint is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.230.** With reference to Angle of Elevation and Depression, which of the following statements is correct?
 A) The foundational proposition regarding Angle of Elevation and Depression is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.231.** With reference to Tangent at any Point of Circle, which of the following statements is correct?
 A) The foundational proposition regarding Tangent at any Point of Circle is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.232.** With reference to Empirical Probability, which of the following statements is correct?
 A) The foundational proposition regarding Empirical Probability is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.233.** With reference to Geometrical Meaning of Zeroes, which of the following statements is correct?
 A) The foundational proposition regarding Geometrical Meaning of Zeroes is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.234.** With reference to Substitution & Elimination, which of the following statements is correct?
 A) The foundational proposition regarding Substitution & Elimination is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.235.** With reference to Quadratic Formula & Nature of Roots, which of the following statements is correct?
 A) The foundational proposition regarding Quadratic Formula & Nature of Roots is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.236.** With reference to General nth Term of an AP, which of the following statements is correct?
 A) The foundational proposition regarding General nth Term of an AP is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.237.** With reference to Area of a Triangle, which of the following statements is correct?
 A) The foundational proposition regarding Area of a Triangle is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above
- Q.238.** With reference to Trigonometric Ratios & Identities, which of the following statements is correct?
 A) The foundational proposition regarding Trigonometric Ratios & Identities is mathematically/factually/practically/empirically valid under standard board conditions
 B) Only secondary/typical derivatives hold true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence/none of the above

- Q.239.** With reference to Length of Tangents from External Point, which of the following statements is correct?
 A) The foundational proposition regarding Length of Tangents from External Point is canonically valid under standard board conditions
 B) Only the secondary observation holds true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence
 D) None of the above
- Q.240.** With reference to Complementary Events, which of the following statements is correct?
 A) The foundational proposition regarding Complementary Events is canonically valid under standard board conditions
 B) Only the secondary observation holds true under standard board conditions
 C) Both theoretical statements are contradicted by standard evidence
 D) None of the above
- Q.241.** Which of the following statements concerning Properties of Determinants is correct?
 A) Statement regarding Properties of Determinants is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.242.** Which of the following statements concerning Rolle & Mean Value Theorems is correct?
 A) Statement regarding Rolle & Mean Value Theorems is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.243.** Which of the following statements concerning Integration by Parts & Partial Fractions is correct?
 A) Statement regarding Integration by Parts & Partial Fractions is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.244.** Which of the following statements concerning Direction Cosines & Shortest Distance is correct?
 A) Statement regarding Direction Cosines & Shortest Distance is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.245.** Which of the following statements concerning Matrix Multiplication & Transpose is correct?
 A) Statement regarding Matrix Multiplication & Transpose is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.246.** Which of the following statements concerning Logarithmic Differentiation is correct?
 A) Statement regarding Logarithmic Differentiation is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.247.** Which of the following statements concerning Separable Variable Differential Equations is correct?
 A) Statement regarding Separable Variable Differential Equations is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.248.** Which of the following statements concerning Dot and Cross Products of Vectors is correct?
 A) Statement regarding Dot and Cross Products of Vectors is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.249.** Which of the following statements concerning Inverse of Matrix & Linear Systems is correct?
 A) Statement regarding Inverse of Matrix & Linear Systems is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.250.** Which of the following statements concerning Chain Rule & Implicit Differentiation is correct?
 A) Statement regarding Chain Rule & Implicit Differentiation is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.251.** Which of the following statements concerning Definite Integral Properties is correct?
 A) Statement regarding Definite Integral Properties is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.252.** Which of the following statements concerning Equation of a Plane is correct?
 A) Statement regarding Equation of a Plane is canonically valid under standard syllabus specifications
 B) Only the secondary observation holds true under standard syllabus specifications
 C) Both statements are factually invalid
 D) None of the above
- Q.253.** "Equivalence Relations" 'YG "‰ 'É 'r .y '‰? 'ém "%2"ù "ù\$ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 [English]: Which of the following statements correctly articulates the core principle of "Equivalence Relations"?
 A) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 B) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 C) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 D) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
- Q.254.** "Invertible Matrices & Adjoint" 'YG "‰ 'É 'r .y '‰? 'ém "%2"ù "ù\$ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 [English]: Which of the following statements correctly articulates the core principle of "Invertible Matrices & Adjoint"?
 A) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 B) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 C) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð
 D) 'ù9 'Y‰', \$W V—'alence Relations" 'YG 'éB")-")\$ "‰? 'iM'y>•) \$BR '(9/ 'éG•" 8"r "É(-"‰> 'Y‰', 8'IM'ò 9" f ð

Q.255. "Second Order Derivatives" 'YG "‰ 'É 'r ."y '‰?'éM'‰2"ù "ù\$ 'éG•" 8"r "É(-"‰> 'Y%', 8'IM'ò 9"fđ

[English]: Which of the following statements correctly articulates the core principle of "Second Order Derivatives"?

[illegible]

Q.256. "Rate of Change of Quantities" 'YG "‰ 'É 'r."y '‰? 'éM'‰2'ù "ù\$ 'éG." 8"r "E(-"‰> 'Y‰, 8'IM'ò 9" fđ

[English]: Which of the following statements correctly articulates the core principle of "Rate of Change of Quantities"?

C) 'ù9 'Y%,' *''U0'É''ù '">%o>YM'YM'UK.' &'U5'é0'à "á" 7''ù ' %9'D' 6''ù 'Y%'>L'UK ubi ubi Eav RhoB statement word tradic'is ENB Prcale U'cein Bsta

Q.257. "Partial Fractions Method" 'YG "%o 'É 'r ."y '%o?'éM'%o2'ü "ü\$ 'éG." 8'f."É (-"%o> 'Y%, 8'IM'ò 9'f'ò

[English]: Which of the following statements correctly articulates the core principle of "Partial Fractions Method"?

A) "ü9 'Y%', % artial Fractions Method" ('YG'éB")-"\$ "%0">im'y>)"SBR(Ü9/YG'Y2 Ü9/GC3)*""%Gm%??'B'ÖR'B."y\$ "A'QöD?TRIS)StalChenPöfura
Method"
C) ü9 'Y%', *""Ü0'eP''ü "%0">YM'YM'ÜK.'" &'Ü5"e0"a *"'7"Ü '%09'D' öH TCG'Y2edUY'ZHDUDTHS statementAdmittations empirical spectroscopic

Q.258. "Linear First Order Differential Equations" 'YG "%o 'É 'r."y "%o? 'eM "%o2 "ù "ù\$ 'éG• "8'r "É(-"%o> 'Y%', 8'IM'ò 9")fđ

[English]: Which of the following statements correctly articulates the core principle of "Linear First Order Differential Equations"?

[illegible]

Q.259. "Dot and Cross Products" 'YG "‰ 'É 'r."y '‰?'éM'‰2"ù "ù\$ 'éG• 8"r "E(-"‰> 'Y%, 8'IM'ò 9"fđ

[English]: Which of the following statements correctly articulates the core principle of "Dot and Cross Products"?

C) 'ù9 'Y%', *'U0'é/'1 "ù "%>'YM'yM'ùK•' &'"U5'e0'a *'7"U '0009" "ù9'Y%Y2edUyL2ZH'U0>"H states fundamental empirical equation
formulates the fundamental governing law of "Dot and Cross Products"
to the primary governing process.

Q.260. "Bayes Theorem Applications" $\text{Y} \cdot \text{G} \cdot \frac{1}{100} \cdot \text{É} \cdot \text{r} \cdot \text{y} \cdot \frac{1}{100} \cdot \text{?} \cdot \text{É} \cdot \text{M} \cdot \frac{1}{100} \cdot \text{2} \cdot \text{ù} \cdot \text{ù} \cdot \text{S} \cdot \text{é} \cdot \text{G} \cdot \text{8} \cdot \text{r} \cdot \text{E} \cdot \left(- \frac{1}{100} \cdot \text{?} \cdot \text{Y} \cdot \frac{1}{100} \cdot \text{8} \cdot \text{IM} \cdot \text{ò} \cdot \text{9} \right) \cdot \text{f} \cdot \text{d}$

[English]: Which of the following statements correctly articulates the core principle of "Bayes Theorem Applications"?

A) 'ù9 'Y%', *''Ù0'é''1' "ù "‰>'YM'yM'ùK.'" & 'ù5'é0'a *'' 7'ù "‰9'B)'ù9'YG'Y2é'Y0'G'W'-'B'‰. This statement contradicts Empirical experiments
standard boundary states.
C) 'ù9 'Y%', \$& yes Theorem Applications" 'YG 'éB'-'-)"\$ "‰>'iM'yB'‰)S'ùK' (G'az 'ùB'Y0'Z'ùP'ùB'‰, \$'B'-'Cé'BiM'\$ a'ù0'ùM' This statement is
Applications" to the primary governing process.

Q.261. "Principal Value Branches of ITF" 'YG "%o 'É 'r ."y '%o?eM%2"ù "u\$ 'eG• 8"r "É(-"%o> 'Y%', 8'IM'ò 9" fđ

[English]: Which of the following statements correctly articulates the core principle of "Principal Value Branches of ITF"?

A) 'ù9 'YG'Y2 •ù 'lL2 *''U0'Ù>'R 9 , '\$'Y> 'éA'iM'd *''U0'YM' ?'ù>B%w %GEXZUY%'e'r #T%BvYThis uk'lely Arcaukiaf Secodary) ffeT U

C) 'ù9 'Y%' , *''U0'e''' "ù "%>'YM'YM'UK•" &'U5'ë0'a "á " 7"ù '%09 D' qn Yoc'y'elidipar Vabed Brinsstemen F'ong Adets e'n p'ic'AmexpeMerita

Q.262. "Matrix Multiplication & Transpose" 'YG "000 'É 'r ."y '000?eM%2"u "u\$ eG." 8"r "E(-"000> 'Y0%, 8'IM ô 9" fô

[English]: Which of the following statements correctly articulates the core principle of "Matrix Multiplication & Transpose"?

A) "ü9 'Y%', \$Ö G ix Multiplication & Transpose" "YG 'éB")-")\$ %"ö09D) ü9 "YG-YZedü9L2H-ü0Ü9Hr StateñentA ContentionÜ9 Empirical "éQö9Hr Multiplication & Transpose"

Q.263. "Logarithmic Differentiation" 'YG "%o 'É 'r ."y '‰?'éM'‰2"ù "ù\$ 'éG•" 8"r "E(-"‰> 'Y%', 8'IM'ò 9" fđ

[English]: Which of the following statements correctly articulates the core principle of "Logarithmic Differentiation"?

A) 'ù9 'YG'Y2 'Y?'iG'r *' ?%'M'Y?'I?'ùK*' . 'y "TM @ . i 'i?'R 0")* "B0ù9 'Y%' , \$EBB "Tm i'c'Offe'ce'ad'0'0'G'G'0B")9%0% This statement potentially

Q.264. "Maxima and Minima & Optimization" 'YG "‰ 'É 'r ."y "‰? 'èM "‰2"ù "ù\$ 'eG." 8"r "É(-"‰> 'Y‰', 8'IM'ò 9")fð

[English]: Which of the following statements correctly articulates the core principle of "Maxima and Minima & Optimization"?

[illegible]

Q.265. "Integration by Substitution & Parts" 'YG "%o 'É 'r ."y to the primary governing process.

[English]: Which of the following statements correctly articulates the core principle of "Integration by Substitution & Parts"?

C) 'ü9 'Y%', \$-çFvpration by Substitution & Parts' 'YG 'éB')."\$ "Ç" "öyres Z-tuna 2-8UÖR B", %6-ÇAİBÖ? BU'ÖM 9'ad %G'Stated

Q.266. "Variable Separable Method" 'YG "%o 'É 'r ."y '%o?'éM'%o2"ù "ù\$ 'éG." 8"r "E(-"%o> 'Y%', 8'IM'ò 9"fä

[English]: Which of the following statements correctly articulates the core principle of "Variable Separable Method"?

C) 'ù9 'Y%' . **'Ù0'É''' 'ù '%>'YM'YM'ÜK.'" & 'Ü5'É0'a " 7'Ü

Q.267. "Shortest Distance between Skew Lines" 'YG "%o 'É'r

[English]: Which of the following statements correctly articulates the core principle of "Shortest Distance between Skew

A) 'ù9 'Y%', %6†÷ test Distance between Skew Lines" 'YG 'éB")-B) "9? W6Y?2\$YPRiCù'd ? 3W6YR10"Jk%G '%?P?G?7BR?C?A?%Gd>Tye"st

standard boundary states

Q.268. "Conditional Probability & Independence" 'YG "‰ 'É 'r. "y '‰? 'éM'‰2"ù "ù\$ 'éG•" 8"r "É(-"‰‰> 'Y‰, 8'IM'ò 9"fð
[English]: Which of the following statements correctly articulates the core principle of "Conditional Probability & Independence"?

A) 'ù9 'YG'Y2 'Y?'IG'r *' ?"%M'Y?'I?ùK•" .ly "™@•i'?'R 0)"
B) Gù9>YB, '\$60e FIM F, %60Â U66 & %ME - (GM @ - 2 F W d / a F i v e 6 R e m e r t) 2ùB
C) 'ù9 'Y%', *'Ù0'é'1 "ù "%>'YM'YM'ùK•" &'Ù5'é0'a *'7"Ù '%9

Q.269. "One-one & Onto Functions" 'Yḡ "‰ 'É 'r 'y '‰%?'ÉM'‰%2'u '‰ eḠ' '8'r 'E'-'‰%>'Y'‰, '8'IM'ò 9'fđ
 [English]: Which of the following statements correctly articulates the core principle of "One-one & Onto Functions"?

A) 'ù9 'Y%', *''U0é'1' 'ù '%>'YM'YM'ùK.&'&'U5é0â 's' 7''U '%99'B)'U9'Y0Y2é(Y0C'YH-B'%.This state is not confirmed by the standard boundary states.

C) 'ù9 'Y%', \$0æR06æ r b öçF0 gVæ7F-öç2" 'r.)2'ÜB'B 8'ù& U)æB 5 G0%2rù YL2% 'U0B'æ'8'9,(0Y)~(æ'W0'>'U0'Y'B 0?This state is not confirmed by the primary governing process.

Q.270. "Solving Linear Systems via Matrix Inverse" 'YG "%0 'E'r'.y "%0?eM"%2'u "u\$ eG•" 8'r" "É(-"%> 'Y%', 8'IM'ò 9" fđ
[English]: Which of the following statements correctly articulates the core principle of "Solving Linear Systems via Matrix Inverse"?

A) 'ù9 'Y%', %6ôÇf-æR /E-æV " 7-7FV×2 f- Ö G ix Inverse" 'Y(ë9)-Y\$ Y2?M?Yç)\$ "R %6MAY?718UôR'Y)" *G@i?%?RBDQ?%G\$%9B6d
"Solving Linear Systems via Matrix Inverse".
C) 'ù9 'Y%', 'Uôe' 'U %> 'M YM UK. & "U5'é0'a * " 7"Ü '%9B'Ü "M9 YG-?2e(U'Y'Z H'BDÜ This statement contradicts empirical %6p6æ66
to the primary governing process.

Q.271. "Chain Rule & Implicit Differentiation" 'YG "% É 'r .y %"?ÉM%2"ú "ú\$ eG• 8'r "E(-"%> 'Y%', 8'IM%2 9")fđ
[English]: Which of the following statements correctly articulates the core principle of "Chain Rule & Implicit Differentiation"?

[illegible]

Q.272. "Increasing and Decreasing Functions" 'Y' 'E' 'r' 'u' 'l' 'e' 's' 'f' 'o' 'r' 'D' 'i' 'f' 'f' 'e' 'r' 'e' 'n' 't' 'i' 'a' 't' 'i' 'o' 'n'
[English]: Which of the following statements correctly articulates the core principle of "Increasing and Decreasing Functions"?

C) "increasing and decreasing functions".

Q.273. "Definite Integral Properties" 'YG "% 'É 'r 'y '%?É'M%'2'u 'u\$ 'éG%' 8'r 'E(-%'>'Y%', 8'IM'ò 9')fđ
[English]: Which of the following statements correctly articulates the core principle of "Definite Integral Properties"?

C) 'ù9 'Y%', **'Ù0'é'P1 "ù "%>'YM'yM'ùK•" &"U5'é0â *7"U " %9D1) "H9 IG-Y2ed UYr ZH-U0U0 This statement conditions empirical exp

to the primary governing process.

Q.274. "Order and Degree" 'Yg "‰ 'É 'r.'y '‰%?'ÉM'‰%2'ù "ù\$ eG.# 8'r "E(-‰%> 'Y%, 8'IM 0 9"f ð
[English]: Which of the following statements correctly articulates the core principle of "Order and Degree"?

C) 'ù9 'Y%', \$-&FW" æB FVfree" 'YG 'éB')-")\$ "%0?im'Y>) \$" R (U) ag -hañdñ Ru YL %Gumñ>>B@?, B'Y %&AñMð * Uñs statemñt ã Cñrãtã&

Q.275. "Direction Cosines & Direction Ratios" 'YG "%0'E r 'y "%0?eM "%2U "u\$ eG•" 8r "E(-"%0> 'Y%', 8'IM'0 9"fδ
[English]: Which of the following statements correctly articulates the core principle of "Direction Cosines & Direction Ratios"?

C) 'ù9 'Y%' ,**'Uò'é'm' "ù "%o>'YM'yM'ùK•" &"ù5'éó'à ** 7"ù '%o9D) "ù9 Y%,"**'Uò'é'm' "ù "%o>'YM'yM'ùK•" &"ù5'éó'à ** 7"ù '%o9D) "ù9 Y%,"**'Uò'é'm' "ù "%o>'YM'yM'ùK•" &"ù5'éó'à ** 7"ù '%o9

Q.276. "Probability Distributions" YG "Cosmos & Direction Ratios" [English]: Which of the following statements correctly articulates the core principle of "Probability Distributions"?

C) "09 %%, "U0é'm'1 "ù "%>'YM'ym'Ük."&"'U5'éóâ *'"7"U' '%gD' D) standard boundary states
to the primary governing process.

Q.277. "Real Numbers & Exponents" YG %00'E'r.'y %00?eM%02"u "u\$ eG•" 8'r "E(-%00>Y%, 8'IM0 9"fd
[English]: Which of the following statements correctly articulates the core principle of "Real Numbers & Exponents"?

C) :ù9 ‘Y%’, %&V Â çVÖ&W’2 b W‡ öæVçG2” “r.”)2ÜB B ’8-ù&“üJelU S G672.-úKZ-%“ UOBjE BRq‘üS’yú\$A YND - *UHBYNö This staGenÉ&

Exponents”:

[English]: Which of the following statements correctly articulates the core principle of "Circles and Tangents"?

C) 'ù9 'Y%', '*'U0'è'm'1' 'ù '%>'YM'ym'ùK.&' *'U5'éó'à' *'7''U '%9'D' 'ù9'10%', 'Sé' >6'E'WZBaBtTing states hen Cc édrad's's'empirical'Exp(ineatá formulates the fundamental governing law of "Circles and

[English]: Which of the following statements correctly articulates the core principle of "Probability and Grouped Data"?

Q 280 Which of the following statements correctly articulates the core principle of "Equivalence Relations"?

Q.280: Which of the following statements correctly articulates the core principle of "Equivalence Relations" ?

A) This statement applies only conditionally under non-standard conditions.

B) This statement accurately formulates the fundamental governing law of "Equivalence Relations".

C) This statement contradicts empirical experimental evidence established by secondary effect unrelated to the primary governing process.

- Q.281.** Which of the following statements correctly articulates the core principle of "Invertible Matrices & Adjoint"?
- A) This statement contradicts empirical experimental evidence established by the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Invertible Matrices & Adjoint".
- Q.282.** Which of the following statements correctly articulates the core principle of "Second Order Derivatives"?
- A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.283.** Which of the following statements correctly articulates the core principle of "Rate of Change of Quantities"?
- A) This statement accurately formulates the fundamental governing law of "Rate of Change of Quantities".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.284.** Which of the following statements correctly articulates the core principle of "Partial Fractions Method"?
- A) This statement applies only conditionally under non-standard boundary states.
 B) This statement accurately formulates the fundamental governing law of "Partial Fractions Method".
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.285.** Which of the following statements correctly articulates the core principle of "Linear First Order Differential Equations"?
- A) This statement accurately formulates the fundamental governing law of "Linear First Order Differential Equations".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.286.** Which of the following statements correctly articulates the core principle of "Dot and Cross Products"?
- A) This statement contradicts empirical experimental evidence established by the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Dot and Cross Products".
- Q.287.** Which of the following statements correctly articulates the core principle of "Bayes Theorem Applications"?
- A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.288.** Which of the following statements correctly articulates the core principle of "Principal Value Branches of ITF"?
- A) This statement accurately formulates the fundamental governing law of "Principal Value Branches of ITF".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.289.** Which of the following statements correctly articulates the core principle of "Matrix Multiplication & Transpose"?
- A) This statement applies only conditionally under non-standard boundary states.
 B) This statement accurately formulates the fundamental governing law of "Matrix Multiplication & Transpose".
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.290.** Which of the following statements correctly articulates the core principle of "Logarithmic Differentiation"?
- A) This statement contradicts empirical experimental evidence established by the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Logarithmic Differentiation".
- Q.291.** Which of the following statements correctly articulates the core principle of "Maxima and Minima & Optimization"?
- A) This statement accurately formulates the fundamental governing law of "Maxima and Minima & Optimization".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.292.** Which of the following statements correctly articulates the core principle of "Integration by Substitution & Parts"?
- A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.293.** Which of the following statements correctly articulates the core principle of "Variable Separable Method"?
- A) This statement accurately formulates the fundamental governing law of "Variable Separable Method".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.294.** Which of the following statements correctly articulates the core principle of "Shortest Distance between Skew Lines"?
- A) This statement applies only conditionally under non-standard boundary states.
 B) This statement accurately formulates the fundamental governing law of "Shortest Distance between Skew Lines".
 C) This statement contradicts empirical experimental evidence established by the primary governing process.
- Q.295.** Which of the following statements correctly articulates the core principle of "Conditional Probability & Independence"?
- A) This statement contradicts empirical experimental evidence established by the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Conditional Probability & Independence".

Q.296. Which of the following statements correctly articulates the core principle of "One-one & Onto Functions"?

A) This is merely an auxiliary secondary effect unrelated to the primary statement in question.
B) This statement is only conditionally under non-standard boundary states.
C) This statement contradicts empirical experimental evidence established by the scientific community.
D) This statement accurately formulates the fundamental governing law of "One-one & Onto Functions".

Q.297. Which of the following statements correctly articulates the core principle of "Solving Linear Systems via Matrix Inverse"?

A) This statement accurately formulates the fundamental governing principle of solvent liquid systems and is fully understood by all boundary states.

B) This statement accurately formulates the fundamental governing principle of solvent liquid systems and is fully understood by all boundary states.

C) This statement contradicts empirical experimental evidence established by primary and auxiliary secondary effect unrelated to the primary governing process.

Q.298. Which of the following statements correctly articulates the core principle of "Chain Rule & Implicit Differentiation"?

A) This statement accurately formulates the fundamental governing principle of the Chain Rule of the Implicit Differentiation under non-standard boundary states.

Q.299. Which of the following statements correctly articulates the core principle of "Increasing and Decreasing Functions"?

A) This statement applies only conditionally under non-standard conditions.
B) This statement accurately formulates the fundamental governing law of "Increasing and Decreasing Functions".
C) This statement contradicts empirical experimental evidence established by the primary governing process.

Q.300. Which of the following statements correctly articulates the core principle of "Definite Integral Properties"?

A) This statement contradicts empirical experimental evidence established by the scientific community.
B) This statement applies only conditionally under non-standard boundary states.
C) This statement accurately formulates the fundamental governing laws of the system.
D) This statement is a secondary effect unrelated to the primary governing process.

Q.301. Which of the following statements correctly articulates the core principle of "Order and Degree"?

A) This is merely an auxiliary secondary effect unrelated to the primary governing principle
B) This statement applies only conditionally under non-standard boundary states
C) This statement contradicts empirical experimental evidence established by the law of "Order and Degree"

Q.302. Which of the following statements correctly articulates the core principle of "Direction Cosines & Direction Ratios"?

A) This statement accurately formulates the fundamental governing principle of the system in a precise and directionally unambiguous manner.

Q.303. Which of the following statements correctly articulates the core principle of "Probability Distributions"?

A) This statement accurately formulates the fundamental governing process of the system. B) This statement is a boundary condition. C) This statement contradicts empirical experimental evidence established in the past. D) This statement is an auxiliary secondary effect unrelated to the primary governing process.

Q.304. Which of the following statements correctly articulates the core principle of "Real Numbers & Exponents"?

A) This is merely an auxiliary secondary effect unrelated to the primary statement principles only conditionally under non-standard boundary states.
 B) This statement contradicts empirical experimental evidence established by the scientific community.
 C) This statement accurately formulates the fundamental governing law of "Real Numbers & Exponents".

Q.305. Which of the following statements correctly articulates the core principle of "Circles and Tangents"?

A) This statement accurately formulates the fundamental governing principle of the system and is not conditional under non-standard boundary states.

B) This statement is not accurate, as it is conditional under non-standard boundary states.

C) This statement contradicts empirical experimental evidence established by the primary governing process.

D) This statement is not accurate, as it is conditional under non-standard boundary states.

Q.306. Which of the following statements correctly articulates the core principle of "Probability and Grouped Data"?

A) This statement applies only conditionally under non-standard conditions.
B) This statement accurately formulates the fundamental governing law of "Probability and Grouped Data".
C) This statement contradicts empirical experimental evidence established by the board of auxiliary secondary effect unrelated to the primary governing process.

Q.307. "Fundamental Theorem of Arithmetic" 'YG "% 'É 'r ."y "%? 'éM "%2 'ù 'ù\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9")fđ

[English]: Which of the following statements correctly articulates the core principle of "Fundamental Theorem of Arithmetic"?

A) Every integer greater than 1 is either a prime number or can be represented as a product of prime numbers.

B) The sum of two prime numbers is always a prime number.

C) Every integer greater than 1 has a unique prime factorization, up to the order of the factors.

D) The product of two prime numbers is always a prime number.

accurately formulates the fundamental governing law of

308. "Substitution and Elimination Methods" 'Yǒu 'Ér yí Fúfān Mùtiáo de Yìngyòng.'
[English]: Which of the following statements correctly articulates the core principle of "Substitution and Elimination Methods"?
A) 'Yǒu 'Y%, %7V'7F—GWF—ôâ æB VÆ-Ö-æ F-ôâ ÖWFtöG2' B) 'Dè VB B'S'Ue U'e B S%'M?YKl%...' C) @·i('ÖY'Ü\$ %

C) "Substitution and Elimination Methods" standard boundary states.

Q.309. "nth Term and Sum of First n Terms of AP" 'YG "%0 É r . y %0? eM %02"u "d \$ eG." 8"r "É(-"%0> 'Y%, 8'IM"0 9" fđ
[English]: Which of the following statements correctly articulates the core principle of "nth Term and Sum of First n Terms of AP"?

[illegible]

Q.310. "Basic Proportionality Theorem (Thales)" 'YG "‰ 'É' r 'y '‰? 'ém'‰2'ù "ù\$ 'éG•" 8'r "É(-'‰> 'Y%', 8'IM'ò 9')/đ

[English]: Which of the following statements correctly articulates the core principle of "Basic Proportionality Theorem (Thales)"?

A) 'ù9 'Y%', **'U0'é'm' 'ù •Y%'>" '%Him'y>)'\$ù " %>'YM'yM'uK'B)&ù95'e'a'z'"Y'üG%'9' ?"%MH'YO'I'Y'ük'"Uly TM'H-@ ò THIR STATEHEG"polyB

C) 'ù9 'Y%', \$ & 6-2 &÷ ÷ tionality Theorem (Thales)" 'YG 'éB')-")'ù9 m'g>+P+r yLza' U0U'FRG,"\$ %G&N'B'U0B'M'\$&S'%dC This é&

Proportionality Theorem (Thales)".

to the primary governing process.

- Q.311.** "Trigonometric Identities ($\sin^2 + \cos^2 = 1$)" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Trigonometric Identities ($\sin^2 + \cos^2 = 1$)"?
- A) 'ù9 'YG"Y2 •ù 'yL'2 "ú0"ù>"R 9", \$Y> 'éA'IM'ò "ú0"YM" ?ù>B%>B%GÉ2ÚY%G"r "TM%>H%BYT%R%ik%ierly%h%au%k%if%a%y%e%e%t%U%
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.312.** "Perimeter and Combinations of Plane Figures" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Perimeter and Combinations of Plane Figures"?
- A) 'ù9 'Y%', % erimeter and Combinations of Plane Figures" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.313.** "Surface Area and Volume of Cylinder & Cone" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Surface Area and Volume of Cylinder & Cone"?
- A) 'ù9 'Y%', %7W&`ace Area and Volume of Cylinder & Cone" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.314.** "Mode and Median of Grouped Data" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Mode and Median of Grouped Data"?
- A) 'ù9 'YG"Y2 "Y?"iG"r " " ?"%M'Y?"I?"ùK•" .y "TM@•i?"R 0")* "%G%>Y%,"\$00FR"HB"ú0"U"á0%W%W"V%G•F9"%d")2%is%Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.315.** "Zeroes of Quadratic Polynomials" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Zeroes of Quadratic Polynomials"?
- A) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', %W&öW2 öb V G atic Polynomials" 'YG 'éB")-)"\$%>Y%,"\$00FR"HB"ú0"U"á0%W%W"V%G•F9"%d")2%is%Statiéa%G%úat%á%&ú
- Q.316.** "Graphical Solution Method" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Graphical Solution Method"?
- A) 'ù9 'YG"Y2 •ù 'yL'2 "ú0"ù>"R 9", \$Y> 'éA'IM'ò "ú0"YM" ?ù>B%>B%GÉ2ÚY%G"r "TM%>H%BYT%R%ik%ierly%h%au%k%if%a%y%e%e%t%U%
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.317.** "Nature of Roots" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Nature of Roots"?
- A) 'ù9 'Y%', \$æ GW&R öb &ö+G2" "r .)2'ÜB'B 8"ú&"Ü"é B 5 %B)ú1YK%G%2 "Y"úG"r ("ú0%M%\$%M%L%K%TM%Hy-B"ú0%is%Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.318.** "Distance Formula and Section Formula" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Distance Formula and Section Formula"?
- A) 'ù9 'Y%', \$F—7F æ6R `ormula and Section Formula" 'YG 'éB")ú1YK%G%2 "Y"úG"r ("ú0%M%\$%M%L%K%TM%Hy-B"ú0%is%Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.319.** "Trigonometric Ratios of Specific Angles" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Trigonometric Ratios of Specific Angles"?
- A) 'ù9 'YG"Y2 "Y?"iG"r " " ?"%M'Y?"I?"ùK•" .y "TM@•i?"R 0")* "%G%>Y%,"\$00FR"HB"ú0"U"á0%W%W"V%G•F9"%d")2%is%Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.320.** "Area of Sector and Segment of a Circle" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Area of Sector and Segment of a Circle"?
- A) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', \$ &V öb 6V7F÷" æB 6VvÖVçB öb 6—&6ÆR" "r .)2'ÜB'B 8"ú&"Ü"é B 2 %B)ú1YK%G%2 "Y"úG"r ("ú0%M%\$%M%L%K%TM%Hy-B"ú0%is%Statiéa%G%úat%á%&ú
- Q.321.** "Conversion of Solids from One Shape to Another" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Conversion of Solids from One Shape to Another"?
- A) 'ù9 'YG"Y2 •ù 'yL'2 "ú0"ù>"R 9", \$Y> 'éA'IM'ò "ú0"YM" ?ù>B%>B%GÉ2ÚY%G"r "TM%>H%BYT%R%ik%ierly%h%au%k%if%a%y%e%e%t%U%
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú
- Q.322.** "Mean of Grouped Data (Direct & Assumed Mean)" 'YG "% É 'r .y %?'ém%2"ú "ú\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9" fð
[English]: Which of the following statements correctly articulates the core principle of "Mean of Grouped Data (Direct & Assumed Mean)"?
- A) 'ù9 'Y%', \$ÖV â öb w&÷W VB F F „F—&V7B b 77VÖVB ÖV æ) 'ù9"Y2G%2"Y"úG"r ("ú0%M%\$%M%L%K%TM%Hy-B"ú0%is%Statiéa%G%úat%á%&ú
C) 'ù9 'Y%', "ú0"é"1 "ú •Y%"Y> "%H'IM'y>•)"\$ù "%>'YM"yM'ùKĐ)ú051éB"á2"ú YL2ú9U0"U"R0Y%\$&("úA"IMHéB"ú0"IM"Statiéa%G%úat%á%&ú

Q.323. "Irrationality Proofs" 'YG "% É 'r ."y '%?'éM'%2"ù "ù\$ 'éG•" 8"r "É(-"%> 'Y%', 8'IM'ò 9")fđ

[English]: Which of the following statements correctly articulates the core principle of "Irrationality Proofs"?

A) '09 'YG'Y2 'Y?'iG'r *' ?'%'M'Y'?!'uK•' . 'y '™@ • i'?'R 0')* %BGu9 YB, \$K-I rationality%00use>%GMB 0')™@%9 May>7)SIS (a)ent appd

C) 'ù9 'Y%, **'Ü0'ê''1 "ù •Y%>'Y> %'H'im'y>.)\$ "ù %>'YM"yM'ùkD)ç05'ê5'â2 •D'YL 2.9'Ü0'Ü'PR0,'Y\$ (P'ÜA'IM'EB'Ü0'M'Statements'G'ö'ü'ê'â
formulates the fundamental governing law of "Irrationality Roots".
to the primary governing process.

Q.324. "Consistency Conditions for Linear Systems" 'YG "‰ 'E r .y '‰?ēM‰2'ū "ū\$ 'ēG•" 8'r "É(-"‰> 'Y‰', 8'IM'ò 9'fđ

[English]: Which of the following statements correctly articulates the core principle of "Consistency Conditions for Linear Systems"?

A) 'ù9 'Y%', \$60ç6—7Fvæ7' 6øæF—F—öç2 `or Linear Systems" "BGÜS'YG\$2%?1NG7-7'S?PqMUYA'I?'8K@.R'0T)*@oGi%R0B'C%G "\$)gB"%
"Consistency Conditions for Linear Systems" standard boundary states. ຈະມີການປັບປຸງເພື່ອໃຫ້ໄດ້ກົງກັນຂອງສາຍຕໍ່ລະຫວ່າງ

C) us Y%, "UO e% u .Y% Y> %HM y.)\$“ü “%>'YM“yM üKd&u059e5 az “ÜYLZ%9 UO“HRO‘,Y\$(PÜATMH%BÜ“MMStatemeGöatr&R

Q.325. "Quadratic Formula & Discriminant" 'YG "% É 'r .y "%? éM "%2 "ú "ù\$ 'éG • 8"r "E(- "%> 'Y%", 8'IM'ò 9" fđ

[English]: Which of the following statements correctly articulates the core principle of "Quadratic Formula & Discriminant"?

A) 'ù9 'Y%' , **U0'6'1" 'ù •Y%'>' %0H'i'm'y>)'\$'ù %0>'YM'yM'ùK8)&09Y6'82""7'U'G%'9'?'%0M'Y0'Y'ùK'U'yTMH@0"THIS state that"coyBaTM

C) 'ù9 'Y%', % V G atic Formula & Discriminant" ('YG 'éB)-")\$ "%0 Divi9-5 \$ R2 a r a L 2 t u d i n e r % 9 % \$ % 2 A B i D B U 0 a i n % d > This state is to the primary governing process.

Q.326. "Criteria for Similarity of Triangles" YG "‰ 'É' r ."y '‰? 'éM'‰2"û "u\$ 'éG•"8"r "É(-"‰> 'Y‰, 8'IM'ò 9" fđ

[English]: Which of the following statements correctly articulates the core principle of "Criteria for Similarity of Triangles"?

[illegible]

accurately formulates the fundamental governing law of "Criteria for

Q.327. "Heights and Distances (Angle of Elevation/Depression)"

[English]: Which of the following statements correctly articulates the core principle of "Heights and Distances (Angle of Elevation/Depression)"?

A) 'ü9 'Y%', \$†V-†tG2 æB F-7F æ6W2 „ ævÆR öb VÆPvations BDepresGörz "YGiGB"-J\$%M?MI9-üJ\$ "Ry(ü76@-8'IDRÖY)"%GG-%B"B" governing law of "Heights and Distances (Angle of Elevation/Depression) of boundary states. <https://www.gutenberg.org/files/54481/54481-h/54481-h.htm>

to the primary governing process.

Q.328. "Tangents to a Circle from an External Point" 'YG "% 'E' r.'y '%? 'eM' %2' u' u\$ 'eG'• 8'r 'É(-"%> 'Y%', 8'IM'ò 9" fđ

[English]: Which of the following statements correctly articulates the core principle of "Tangents to a Circle from an External Point"?

A) 'u9 'YG"Y2 "Y?"iG"r "*" ?"%M'Y?!"uK•" .!y "™@•! "i?"R 0")* "BGu9 Y2 "™%@anghts%"a "Üde %müa ("Ex"erha"Print"MS"stär"neß"appli
statement accurately formulates the fundamental governing law of

C) 'ù9 'Y%' , **'U0'é'm' 'ù •Y%'>' %Him'y>•)\$'ù '%>'YM'yM'uKp

Q.329. "Volume of Sphere and Hemisphere" 'YG "‰ 'É 'r ."y '‰?'éM'‰2"ú "ù\$ 'éG•" 8"r "E(-"‰> 'Y%, 8'IM'ò 9" /đ

[English]: Which of the following statements correctly articulates the core principle of "Volume of Sphere and Hemisphere"?

A) 'ù9 'Y%' , **U0'è/'1 'ù •Y%'>' %0H'iM'y>.)\$'ù '%>'YM'yM'ùK8)&095Y6'32'''7'U'G%'9'?'%TM'Y0'Y'ùK('U'y TM'H@ 0 THIS STATE'neat'c'p'è'at'D

C) 'ù9 'Y%', %òlume of Sphere and Hemisphere" ('G éB")-)"\$ (6) Nv>S2R-DU 2 8 U@>R)9%\$G-%A B0?BJ0 \$1d 9%ed %0S "st&d
of Sphere and Hemisphere".
to the primary governing process.

Q.330. "Classical Definition of Probability & Simple Events"

[English]: Which of the following statements correctly articulates the core principle of "Classical Definition of Probability & Simple Events"?

law of "Classical Definition of Probability & Simple Events" standard boundary states

to the primary governing process.

Q.331. Which of the following statements correctly articulates the core principle of "Fundamental Theorem of Arithmetic"?

A) This statement accurately formulates the fundamental governing principle of the first fundamental theorem of thermodynamics.
B) This statement accurately formulates the fundamental governing principle of the second fundamental theorem of thermodynamics.
C) This statement contradicts empirical experimental evidence concerning the thermodynamic boundary states.
D) This statement accurately formulates the fundamental governing principle of the third fundamental theorem of thermodynamics.

C) This statement contradicts empirical experimental evidence established by the board. auxiliary secondary effect unrelated to the primary governing process.

Q.332. Which of the following statements correctly articulates the core principle of "Substitution and Elimination Methods"?

A) This statement applies only conditionally under non-standard conditions. B) This statement accurately formulates the fundamental governing law of "Substitution and Elimination Methods".

C) This statement contradicts empirical experimental evidence established by the primary governing process.

Q.333. Which of the following statements correctly articulates the core principle of "nth Term and Sum of First n Terms of AP"?

A) This statement contradicts empirical experimental evidence established by the scientific community. B) This statement applies only conditionally under non-standard boundary states.

C) This statement accurately formulates the fundamental governing law of the international system of states.

Q.334. Which of the following statements correctly articulates the core principle of "Basic Proportionality Theorem (Thales)"?

A) This is merely an auxiliary secondary effect unrelated to the primary steering properties.
B) This is steering properties only conditionally under non-standard boundary states.

C) This statement contradicts empirical experimental evidence established by the boundary states law of "Basic Proportionality Theorem (Thales)".

- Q.335.** Which of the following statements correctly articulates the core principle of "Trigonometric Identities ($\sin^2 + \cos^2 = 1$)"?
 A) This statement accurately formulates the fundamental governing law of "Trigonometric Identities ($\sin^2 + \cos^2 = 1$)".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.336.** Which of the following statements correctly articulates the core principle of "Perimeter and Combinations of Plane Figures"?
 A) This statement accurately formulates the fundamental governing law of "Perimeter and Combinations of Plane Figures".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.337.** Which of the following statements correctly articulates the core principle of "Surface Area and Volume of Cylinder & Cone"?
 A) This statement applies only conditionally under non-standard boundary states.
 B) This statement accurately formulates the fundamental governing law of "Surface Area and Volume of Cylinder & Cone".
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.338.** Which of the following statements correctly articulates the core principle of "Mode and Median of Grouped Data"?
 A) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Mode and Median of Grouped Data".
- Q.339.** Which of the following statements correctly articulates the core principle of "Zeroes of Quadratic Polynomials"?
 A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.340.** Which of the following statements correctly articulates the core principle of "Graphical Solution Method"?
 A) This statement accurately formulates the fundamental governing law of "Graphical Solution Method".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.341.** Which of the following statements correctly articulates the core principle of "Nature of Roots"?
 A) This statement applies only conditionally under non-standard boundary states.
 B) This statement accurately formulates the fundamental governing law of "Nature of Roots".
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.342.** Which of the following statements correctly articulates the core principle of "Distance Formula and Section Formula"?
 A) This statement accurately formulates the fundamental governing law of "Distance Formula and Section Formula".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.343.** Which of the following statements correctly articulates the core principle of "Trigonometric Ratios of Specific Angles"?
 A) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Trigonometric Ratios of Specific Angles".
- Q.344.** Which of the following statements correctly articulates the core principle of "Area of Sector and Segment of a Circle"?
 A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.345.** Which of the following statements correctly articulates the core principle of "Conversion of Solids from One Shape to Another"?
 A) This statement accurately formulates the fundamental governing law of "Conversion of Solids from One Shape to Another".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.346.** Which of the following statements correctly articulates the core principle of "Mean of Grouped Data (Direct & Assumed Mean)"?
 A) This statement applies only conditionally under non-standard boundary states.
 B) This statement accurately formulates the fundamental governing law of "Mean of Grouped Data (Direct & Assumed Mean)".
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
- Q.347.** Which of the following statements correctly articulates the core principle of "Irrationality Proofs"?
 A) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement accurately formulates the fundamental governing law of "Irrationality Proofs".
- Q.348.** Which of the following statements correctly articulates the core principle of "Consistency Conditions for Linear Systems"?
 A) This statement accurately formulates the fundamental governing law of "Consistency Conditions for Linear Systems".
 B) This statement applies only conditionally under non-standard boundary states.
 C) This statement contradicts empirical experimental evidence established by the board auxiliary secondary effect unrelated to the primary governing process.

Q.349. Which of the following statements correctly articulates the core principle of "Quadratic Formula & Discriminant"?

- A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
B) This statement applies only conditionally under non-standard boundary states.
C) This statement contradicts empirical experimental evidence established by the board.
D) This statement accurately formulates the fundamental governing law of "Quadratic Formula & Discriminant".

Q.350. Which of the following statements correctly articulates the core principle of "Criteria for Similarity of Triangles"?

- A) This statement accurately formulates the fundamental governing law of "Criteria for Similarity of Triangles".
B) This statement applies only conditionally under non-standard boundary states.
C) This statement contradicts empirical experimental evidence established by the board.
D) This statement is merely an auxiliary secondary effect unrelated to the primary governing process.

Q.351. Which of the following statements correctly articulates the core principle of "Heights and Distances (Angle of Elevation/Depression)"?

- A) This statement applies only conditionally under non-standard boundary states.
B) This statement accurately formulates the fundamental governing law of "Heights and Distances (Angle of Elevation/Depression)".
C) This statement contradicts empirical experimental evidence established by the board.
D) This statement is merely an auxiliary secondary effect unrelated to the primary governing process.

Q.352. Which of the following statements correctly articulates the core principle of "Tangents to a Circle from an External Point"?

- A) This statement contradicts empirical experimental evidence established by the board.
B) This statement applies only conditionally under non-standard boundary states.
C) This statement accurately formulates the fundamental governing law of "Tangents to a Circle from an External Point".
D) This statement is merely an auxiliary secondary effect unrelated to the primary governing process.

Q.353. Which of the following statements correctly articulates the core principle of "Volume of Sphere and Hemisphere"?

- A) This is merely an auxiliary secondary effect unrelated to the primary governing process.
B) This statement applies only conditionally under non-standard boundary states.
C) This statement contradicts empirical experimental evidence established by the board.
D) This statement accurately formulates the fundamental governing law of "Volume of Sphere and Hemisphere".

Q.354. Which of the following statements correctly articulates the core principle of "Classical Definition of Probability & Simple Events"?

- A) This statement accurately formulates the fundamental governing law of "Classical Definition of Probability & Simple Events".
B) This statement applies only conditionally under non-standard boundary states.
C) This statement contradicts empirical experimental evidence established by the board.
D) This statement is merely an auxiliary secondary effect unrelated to the primary governing process.

Q.355. BODMAS '%@?'. '%G "™2 'Y@'É?•ó 24 / 4 + 3 * (8 - 5)

[English]: Evaluate using BODMAS: $24 / 4 + 3 * (8 - 5)$

- A) 15
B) 27
C) 12
D) 18

Q.356. 'iM"Y?"%>'B 8'é@'Y0'2 ...ã" Ò w,² " Ò "r.")2 'ÉM'é>'B " "ù -@

[English]: Find the roots of the quadratic equation: $x^2 - 7x + 12 = 0$.

- A) -3, -4
B) 3, 4
C) 2, 6
D) 1, 12

Q.357. A 'Y?'%@ 'Y>' M'ò "² &"ù('éG•" \$'Y> B •™8"r R &"ù('éG•" *)0"â " \$"â 9"%d 'iK'%K•" . "ù2'Y0 •™8 'Y>' M'ò "² "ù\$'%G

[English]: A can complete a piece of work in 10 days and B in 15 days. Working together, how many days will they take?

- A) 5 'i?', ò R F ys
B) 6 'i?', ò b F ys
C) 7.5 'i?', ò rãR F ys
D) 8 'i?', ò , F ys

Q.358. sin 30° + cos 60° 'Y> 'é>', "Ù/"â 9"¹ "ãð

[English]: What is the value of $\sin 30^\circ + \cos 60^\circ$?

- A) 0
B) 1/2
C) " 3 / sqrt(3)
D) 1

Q.359. '©M" %'â ""é '¢ 'Ù>'ÉM'ò 8•) "Ù/'é •" "â . "é""Ù/ 'ÉM'é>'B " "ù -@

[English]: Find the mean of the first five prime numbers.

- A) 5.6
B) 5.2
C) 4.8
D) 6.0

Q.360. •ù "Y0"Ù 'Y> '©0"ù. "é* 48 "%G'é@ "™H-B "% "â "Ù7"y\$"Ù0'12 'YM'ù> "™K'y>?

[English]: A square has a perimeter of 48 cm. What is its area?

- A) 196 "%G'é@² / 196 cm²
B) 144 "%G'é@² / 144 cm²
C) 120 "%G'é@² / 120 cm²
D) 169 "%G'é@² / 169 cm²

Q.361. "%0"" " "ù : (0.3 * 0.3 + 0.01) / 0.1

[English]: Simplify: $(0.3 * 0.3 + 0.01) / 0.1$

- A) 1.0
B) 0.1
C) 0.9
D) 10.0

Q.362. •Y 'Y'9?'B " 8•) "Ù/"é\$"Ù. 'R 'Ù?'YM"y.'l> '©0 •" "é0"ù\$ '©M" >'é>9?'R *""Ù0'iM'f "%9" 5"ù ")M'¢ "â "ù('Y@'É?•ó

[English]: Evaluate the following standard railway exam problem on Commercial Arithmetic & Quantitative Methods: Which choice represents the correct outcome?

- A) 'é>% "Y?'Y2"Ù* 1 / Standard Outcome 1
B) '©M" . "é#"ù\$ "™2 "Y?'Y2"Ù* 2 / Verified Solution Criterion 2
C) '©M" >'Y. "ù "%0?"iM'y>•)\$ 3 / Primary Canonical Assertion 3
D) •Y("Ù/ 'Y>" 4 / Auxiliary Factor 4

Q.376. 'ù&"ò B = 2 : 3 'l%"â " C = 4 : 5 "™K, 'IK A : B : C 'ÉM'é>'B " "ù –@

[English]: If $A : B = 2 : 3$ and $B : C = 4 : 5$, find $A : B : C$.

A) 8 : 12 : 15

B) 6 : 9 : 10

C) 2 : 7 : 5

D) 8 : 10 : 15

Q.377. " A 2,000 '©0 5% "Y>" M"y?'R &" 8"r 2 5" M"r "â 8"é""é0'2 , "Ù/"é 'YM'ù> "™K'y>?

[English]: Calculate the simple interest on Rs. 2,000 at 5% per annum for 3 years.

A) " A 250 / Rs. 250

B) " A 300 / Rs. 300

C) " A 350 / Rs. 350

D) " A 400 / Rs. 400

Q.378. •ù 'ùM" G', B •) "r .y 360 'Y?'é@ 'Y@ 'iB" @ 'l/ 'Y0'l@ "™H–B "%o " "é2 'é@/"%oG 'éG•" "Ù/"â 9" f ð

[English]: A train travels 360 km in 4 hours. What is its speed in m/s?

A) 20 'é@/"%oG / 20 m/s

B) 25 'é@/"%oG / 25 m/s

C) 30 'é@/"%oG / 30 m/s

D) 35 'é@/"%oG / 35 m/s

Q.379. Trigonometric Heights & Distances '©0 •i""é0"ù\$ '©M" 6"Ù(1 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Trigonometric Heights & Distances problem scenario #1?

A) 24

B) 36

C) 48

D) 60

Q.380. Probability & Combinatorics Basics '©0 •i""é0"ù\$ '©M" 6"Ù(2 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Probability & Combinatorics Basics problem scenario #2?

A) 24

B) 36

C) 48

D) 60

Q.381. Circular Cylinder & Cone Surface Area '©0 •i""é0"ù\$ '©M" 6"Ù(3 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Circular Cylinder & Cone Surface Area problem scenario #3?

A) 24

B) 36

C) 48

D) 60

Q.382. Boat and Stream Relative Velocity '©0 •i""é0"ù\$ '©M" 6"Ù(4 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Boat and Stream Relative Velocity problem scenario #4?

A) 24

B) 36

C) 48

D) 60

Q.383. Algebraic Factorization & Polynomials '©0 •i""é0"ù\$ '©M" 6"Ù(5 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Algebraic Factorization & Polynomials problem scenario #5?

A) 24

B) 36

C) 48

D) 60

Q.384. Algebraic Factorization & Polynomials '©0 •i""é0"ù\$ '©M" 6"Ù(10 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Algebraic Factorization & Polynomials problem scenario #10?

A) 24

B) 36

C) 48

D) 60

Q.385. Trigonometric Heights & Distances '©0 •i""é0"ù\$ '©M" 6"Ù(11 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Trigonometric Heights & Distances problem scenario #11?

A) 24

B) 36

C) 48

D) 60

Q.386. Probability & Combinatorics Basics '©0 •i""é0"ù\$ '©M" 6"Ù(12 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Probability & Combinatorics Basics problem scenario #12?

A) 24

B) 36

C) 48

D) 60

Q.387. Circular Cylinder & Cone Surface Area '©0 •i""é0"ù\$ '©M" 6"Ù(13 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Circular Cylinder & Cone Surface Area problem scenario #13?

A) 24

B) 36

C) 48

D) 60

Q.388. Boat and Stream Relative Velocity '©0 •i""é0"ù\$ '©M" 6"Ù(14 'Y> 'é>', "Ù/"â 9"1 "ã ð

[English]: What is the evaluated result for Boat and Stream Relative Velocity problem scenario #14?

A) 24

B) 36

C) 48

D) 60

- Q.389.** Algebraic Factorization & Polynomials '©0 •i""é0"ù\$ '©M" 6"Ù(15 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Algebraic Factorization & Polynomials problem scenario #15?
A) 24 B) 36
C) 48 D) 60
- Q.390.** Trigonometric Heights & Distances '©0 •i""é0"ù\$ '©M" 6"Ù(16 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Trigonometric Heights & Distances problem scenario #16?
A) 24 B) 36
C) 48 D) 60
- Q.391.** Probability & Combinatorics Basics '©0 •i""é0"ù\$ '©M" 6"Ù(17 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Probability & Combinatorics Basics problem scenario #17?
A) 24 B) 36
C) 48 D) 60
- Q.392.** Circular Cylinder & Cone Surface Area '©0 •i""é0"ù\$ '©M" 6"Ù(18 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Circular Cylinder & Cone Surface Area problem scenario #18?
A) 24 B) 36
C) 48 D) 60
- Q.393.** Boat and Stream Relative Velocity '©0 •i""é0"ù\$ '©M" 6"Ù(19 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Boat and Stream Relative Velocity problem scenario #19?
A) 24 B) 36
C) 48 D) 60
- Q.394.** Algebraic Factorization & Polynomials '©0 •i""é0"ù\$ '©M" 6"Ù(20 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Algebraic Factorization & Polynomials problem scenario #20?
A) 24 B) 36
C) 48 D) 60
- Q.395.** Trigonometric Heights & Distances '©0 •i""é0"ù\$ '©M" 6"Ù(21 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Trigonometric Heights & Distances problem scenario #21?
A) 24 B) 36
C) 48 D) 60
- Q.396.** Probability & Combinatorics Basics '©0 •i""é0"ù\$ '©M" 6"Ù(22 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Probability & Combinatorics Basics problem scenario #22?
A) 24 B) 36
C) 48 D) 60
- Q.397.** Circular Cylinder & Cone Surface Area '©0 •i""é0"ù\$ '©M" 6"Ù(23 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Circular Cylinder & Cone Surface Area problem scenario #23?
A) 24 B) 36
C) 48 D) 60
- Q.398.** Boat and Stream Relative Velocity '©0 •i""é0"ù\$ '©M" 6"Ù(24 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Boat and Stream Relative Velocity problem scenario #24?
A) 24 B) 36
C) 48 D) 60
- Q.399.** Algebraic Factorization & Polynomials '©0 •i""é0"ù\$ '©M" 6"Ù(25 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Algebraic Factorization & Polynomials problem scenario #25?
A) 24 B) 36
C) 48 D) 60
- Q.400.** Trigonometric Heights & Distances '©0 •i""é0"ù\$ '©M" 6"Ù(6 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Trigonometric Heights & Distances problem scenario #6?
A) 24 B) 36
C) 48 D) 60
- Q.401.** Probability & Combinatorics Basics '©0 •i""é0"ù\$ '©M" 6"Ù(7 'Y> 'é>', "Ù/"â 9"1 "ãð
[English]: What is the evaluated result for Probability & Combinatorics Basics problem scenario #7?
A) 24 B) 36
C) 48 D) 60

Q.402. Circular Cylinder & Cone Surface Area '©0 •i"é0"ù\$ '©M" 6"Ù(8 'Y> 'é>', "Ù/"à 9"1 "äð
[English]: What is the evaluated result for Circular Cylinder & Cone Surface Area problem scenario #8?

- A) 24 B) 36
C) 48 D) 60

Q.403. Boat and Stream Relative Velocity '©0 •i"é0"ù\$ '©M" 6"Ù(9 'Y> 'é>', "Ù/"à 9"1 "äð
[English]: What is the evaluated result for Boat and Stream Relative Velocity problem scenario #9?

- A) 24 B) 36
C) 48 D) 60

Q.404. Algebra & Polynomials 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'
[English]: In the context of Algebra & Polynomials, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.405. Trigonometry & Heights 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'
[English]: In the context of Trigonometry & Heights, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.406. Geometry & Circles 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9>'
[English]: In the context of Geometry & Circles, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.407. Coordinate Geometry 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9
[English]: In the context of Coordinate Geometry, which of the following options represents the correct scientific / analytical deduction?

- A) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &
C) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver

Q.408. Probability & Statistics 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9
[English]: In the context of Probability & Statistics, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.409. Number Systems & HCF/LCM 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9
[English]: In the context of Number Systems & HCF/LCM, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.410. Time, Speed & Distance 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9
[English]: In the context of Time, Speed & Distance, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.411. Arithmetic & Percentages 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9
[English]: In the context of Arithmetic & Percentages, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

Q.412. Integrals: Definite & Indefinite 'YG "%o 'i0"Ù- 'éG•"Â ("ù."Ù(")?i?'B .y "%oG 'YL', 8"â 5"ù ")M'¢ 8"™@ "YH'ÉM'é>"%o?'Rù5"ù6"Ù2"y7'9
[English]: In the context of Integrals: Definite & Indefinite, which of the following options represents the correct scientific / analytical deduction?

- A) 'Y%', ' 'R 'é>"%oM'ð "Y""é0'9> 'YK 'i0"Ù6"é\$"â 9", ò 7F FVÖVqB) "éV%&W5V'¢200æ>àVàBpùenùeé 'B *)0"Ù#I "%o\$"Ù/ "™H / Standard ver
C) 'iK'%"K•" 'Y('©M" /"1 "é\$"Ù.'R '\$'YM'ùK•" "r 5"ù** @'B 9"%o / BD)H'YKpòsùm(§òhtràçt 2Bpùrùkùsèrvà00as"™@•" " \$"â ò æV—F†W" &

A) 'Y%', 'R' 'é'>%M'd "Y"é0'9> 'YK' i0'U6"é\$ "â'9", ò 7F FVÖV, (B) &W%W5V; G20&e>=a2B1p1en1see 'B *")0'U#1 "%%\$Û/ "™H / Standard ver
C) 'iK'%%K•" 'Y('©M' /"1 "é\$Û.'R '\$YM'ùK•" "r 5"ù* " @'B 9"% / BD)1pK0p0s1m1n1s1o1h1p1d1c1t1B1p1r1m1l1K1o1s1v1a1d1a1s "™@•" " \$â ò æV—F†W" &

A) 'Y%', ' 'R 'é>%M'ò "Y"é0'9> YK i0"U6"é\$â 9",ò 7F FVÖV, (B) &W%W5Vç G00é>=a2B p1eñsée 'B *")0"U#I "%%\$Ù/ "™H / Standard ver
C) 'iK"%K•" 'Y('©M' /"1 "é\$Ù.'R \$YM'ùK•" "r 5"ù* @'B 9"% / BD)H p1o pòs1m1n\$%0h1p1c12B p1r1v1ll1K101s1v1d1s1"™@•" " \$â ò æV—F†W" &

A) 'Y%', ' 'R 'é>%M'ò "Y"é0'9> YK 'i0"U6"é\$â 9",ò 7F FVÖV (B) &V%&W5Vç G200&=a2B1B1p1en1sée 'B *")0"U#1 "%%\$Ü/ "™H / Standard ver
C) 'iK'%%K•" 'Y('©M' /"1 "é\$Ü.'R \$'YM'üK•" "r 5"ü*" @'B 9"% / BD)YK proposition\$%&h1p1d1c1t1B1p1r1f1c1l1o1s1v1a1s "™@•" " \$â ò æV—F†W" &

A) 'Y%', 'R' 'é'>%M'ò "Y"é'0'9> YK 'i'0'U6"é\$à'9",ò 7F FVÖV (B) &V%&W5V'G200ë>2021B'p'ent'isé 'B'*)'0'U#l "%\$%U' / "TMH / Standard ver
C) 'iK'%K•" 'Y('©M' / "1"é\$Ù.R '\$'YM'ùK•" "r 5"ù" @'B'9"% / BD) 'p'rop'osit'ion' (S'ch'w'ap'p'et' B'p'ir'it'u'el'k'ö's'erv'at'io'n'as' "TM @•" " \$"â ò æV—F†W" &

C) 'iK%'K•' 'Y('©M'/'1" "és"Ù.R '\$YM'ùK•' "r 5"ù* @'B 9"% / BD)H)K Propositions (Schwartz & Enfield 2004: 104) " " \$ "à ò æV—F+W" &

C) 'iK%'K•' 'Y ('©M'/'1 "és"Ù.R \$YM'ùK•' "r 5"ù*" @'B 9"% / B)H YKopositiv(§%©hmapct Empirical Observat)as "™ @•" " \$"à ò æV—F↑W" &

C) 'iK'%K•" ")\$ "™H•" ò &÷F, æB " &R W&öæV÷W0 D) •y('éG•" 8"r "1 '%o9" / None of these

Q.420. 'Y2', „6 Æ7VÇW2Â 8i?b „ 'y#“ù\$ •Y%“Y> •i5"Ù"/)9 '©0 •i“é0“ù\$ •™ "Ù 'y#“ù\$ / '©M“ 6"Ù(1 'Y> 'é>', "Ù “é\$ 'Y@'É?•ó

C) "%."é ")(("%%@'é>= pi / 2 / Integral limit= pi / 2 D) 'é>'©>) = " 2 / Modulus= sqrt(2)

Q.421. 'Y2', „6 Æ7VÇW2Â 8'i?b, " 'y#“ù\$ •Y%“Y> •i5”Ù”)9 '©0 •i“é0“ù\$ •™ ”Ù 'y#“ù\$” / '©M“ 6”Ù(2 'Y> 'é>', ”Ù “é\$ 'Y@'É?•ó

C) "%."é")(("%"@'é>= pi / 2 / Integral limit= pi / 2 D) 'é>'©>." = " 2 / Modulus= sqrt(2)

C) "%."é")(("%"@'é'> = pi / 2 / Integral limit = pi / 2 D) 'é>'©>.) = " 2 / Modulus = sqrt(2)

C) "%_."é ") ("%_@'é> = pi / 2 / Integral limit = pi / 2

A) •Y5'Y2', . "é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$""Ù.'R 9"" Ò ò æ Ç—F-6 Â 6öÇWF-ââ Ò
C) "%."é ") ("%@'"é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>•) = " 2 / Modulus = sqrt(2)

A) •Y5'Y2', . "é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$""Ù.'R 9"" Ò ò æ Ç—F-6 Â 6öÇWF-ââ Ò
C) "%."é ") ("%@."é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>•) = " 2 / Modulus = sqrt(2)

A) •Y5'Y2', .é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$"Û.'R 9"" Ò ò æ Ç—F-6 Â 6ôÇWF-ââ Ò
C) "%."é ") ("%"@'é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>•) = " 2 / Modulus = sqrt(2)

A) •Y5'Y2', .é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$"Ù.'R 9"" Ô ò æ Ç—F-6 Â 6ôÇWF-ââ Ô
C) "%,."é ") ("%"@'é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>•) = " 2 / Modulus = sqrt(2)

A) •Y5'Y2', . "é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$""Ù.'R 9"" Ò ò æ Ç—F–6 Â 6öÇWF–ââ Ò
C) "%_."é ") ("%_@'"é>= pi / 2 / Integral limit = pi / 2
D) 'é>'<@>•) = " 2 / Modulus = sqrt(2)

A) •Y5'Y2', .“é(= 0 / Derivative value = 0
B) “Y?”iM”)G“y#“é\$”Ù.'R 9“” Ò ò æ Ç—F–6 Â 6ôÇWF–ââ Ò
C) “%.”é “)(“%@”é>= pi / 2 / Integral limit = pi / 2
D) ‘é>’©>•) = “ 2 / Modulus = sqrt(2)

A) $\bullet Y5'Y2'$, $\therefore \acute{e} = 0$ / Derivative value = 0
B) $\text{"Y?'\imath"}G^{\text{"y\#\acute{e}\$"}\grave{U}.'R\ 9^{uu}}\grave{O}\ \grave{o}\ \text{\ae}\ \text{\c C}—F-6\ \hat{A}\ 6\acute{o}\text{\c C}WF-\grave{o}\grave{o}\ \grave{O}$
C) $\text{"}\%_{oo}.\acute{e}\ \text{"}(\text{"}\%_{oo}\text{@'}\acute{e}>= \pi / 2$ / Integral limit = $\pi / 2$
D) $\acute{e}'>'\text{\c C}>\bullet) = \text{"}^2$ / Modulus = $\text{sart}(2)$

A) •Y5'Y2', "é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$""Ù.'R 9"" Ò ò æ Ç—F-6 Â 6ôÇWF-ââ Ò
C) "%%" "é ") ("%"@'"é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>•) = " 2 / Modulus = sqrt(2)

A) $\nabla Y^5 Y^2$, "é(= 0 / Derivative value = 0

B) $Y^2 \cdot i^m G^y \# \text{é} \$^m \text{Ù} \cdot R^9 m \text{ Ò } \text{ò} \text{ æ } \text{Ç} - 6 \text{ Â } 6 \text{ ö } \text{Ç} \text{ W F} - \text{ä ä } \text{Ò}$

C) $\frac{\%_0}{\%_0} \cdot \text{é} \text{ " } (\frac{\%_0}{\%_0} @ \text{é} > = \text{p j} / 2 / \text{Integral limit} = \text{p j} / 2$

D) $\text{'é} > \text{'@} > \bullet \text{ " } ^2 / \text{Modulus} = \text{sart}(2)$

A) $\bullet Y5'Y2'$, $\cdot \acute{e} = 0$ / Derivative value = 0

B) $\text{"Y?\"iM\"G\"y\#\\"e\$\"}\grave{U}.R\ 9^{u\text{''}}\ \grave{O}\ \grave{o}\ \text{\ae}\ \text{\c C}\text{---}6\ \text{\AA}\ 6\text{\o}\text{\c C}WF\text{---}\text{\aa}\ \grave{O}$

C) $\text{"}\%_{\text{"}}\text{"}\acute{e}\text{"}\text{"}\text{"}\%_{\text{"}}@'\acute{e}>=n_{\text{j}}/2$ / Integral limit = $n_{\text{j}}/2$

D) $\text{"}\acute{e}>\text{"}\text{\textcircled{C}}>\bullet\text{"} = \text{"}^2$ / Modulus = $\text{sart}(2)$

A) "Y5'Y2", "é" = 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$""Ù.'R 9"" Ò ò æ Ç—F 6 Â 6öÇWF—ââ Ò
C) "%"" "é") ("%"@'"é"> = pi / 2 / Integral limit = pi / 2
D) 'é">'@>•) = " 2 / Modulus = sqrt(2)

Q.435. 'Y2', '6 Æ7VÇW2Â 8i?'b , " 'y#"\$ ¥%>'Y> •i5"Ù"/")9 '©0 •i""é0"ù\$ •™ "Ù 'y#"ù\$"/ '©M" 6"Ù(16 'Y> 'é>', "Ù "és 'Y@'É?•0
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #16.

- A) •Y5'Y2', .“é(= 0 / Derivative value = 0
 B) “Y?”iM”)G“y#“é\$”Ù. R 9”” Ò ò æ Ç—F—6 Â 6öÇWF—ôâ Ò
 C) “%.“é“((“%@’é>= pi / 2 / Integral limit = pi / 2
 D) ’é>’©>) = " 2 / Modulus = sqrt(2)

Q.436. 'Y2', „6 Æ7VÇW2Â 8i?"b ,," 'y#“ù\$ •Y%“Y> •i5”Ù”)9 ‘©0 •j““é0“ù\$ •™ ”Ù ‘y#“ù\$” / ‘©M“ 6”Ù(17 ‘Y> ‘é>’, ”Ù “és ‘Y@‘É?•ó
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #17.

- A) •Y5'Y2', "é(= 0 / Derivative value = 0

Q.437. 'Y2', „6 A7VÇW2Â 8i?b „, 'y#“ù\$ •Y%“Y> •i5“Ù”)9 '©0 •i“é0“ù\$ •™ “Ù 'y#“ù\$ / '©M“ 6“Ù(18 'Y> 'é>', “Ù “és 'Y@‘É?•0
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x,
 determine the value / derivative / integral satisfying condition #18.

- A) •Y5'Y2', .´é = 0 / Derivative value = 0
B) "Y?"iM")G"y#`és"Ù. R 9" ¨ ò æ Ç—F—6 Â6öÇWF—öâ Ò
C) "%..´é") ("%"@´é>= pi / 2 / Integral limit = pi / 2
D) ´é>'©>)" = 2 / Modulus = sqrt(2)

Q.438. 'Y2', „6 A7VÇW2Â 8i?‚b „ 'y#“ù\$ •Y%“Y> •i5“Ù”)9 '©0 •i““é0“ù\$ •™ “Ù 'y#“ù\$” / '©M“ 6“Ù(19 'Y> 'é>', “Ù “és 'Y@‘É?•0
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x,
 determine the value / derivative / integral satisfying condition #19.

- A) $\bullet Y^5 Y^2$, $\cdot \acute{e} = 0$ / Derivative value = 0
B) $\text{"Y?"iM"}G\text{"y"}\acute{e}\text{"s"}\grave{U}\cdot R\text{"9"}\grave{O}\ \grave{o}\ \text{æ}\ \text{Ç}—F\text{—}6\ \hat{A}\ \hat{ö}\text{ÇWF}\text{—}\text{öä}\ \grave{O}$
C) $\text{"%."é"}(\text{"%@"}\acute{e}) = \pi / 2$ / Integral limit = $\pi / 2$
D) $\acute{e}'\text{>"}\text{<} \text{"} = 2$ / Modulus = $\sqrt{2}$

Q.439. 'Y2', „6 A7VÇW2Â 8i?\"b ,\" 'y#\"ù\$ •Y%\"Y> •i5\"Ù\"/)9 '©0 •i\"\"é0\"ù\$ •™ \"Ù 'y#\"ù\$\" / '©M\" 6\"Ù(20 'Y> 'é>', \"Ù \"é\$ 'Y@'É?•ó
[English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #20.

- A) •Y5'Y2', "é(= 0 / Derivative Value = 0
B) "Y?"iM")G"y#"é\$".Ù. R 9"" Ò ò æ Ç—F—6 Â6öÇWF—ôâ Ò
C) "%."é")("%"@'é>= pi / 2 / Integral limit = pi / 2
D) 'é>'©>) = " 2 / Modulus = sqrt(2)

Q.440. 'Y2', „6 Æ7VÇW2'Â 8i?"b ,," 'y#"ù\$ •Y%“Y> •i5"Ù"/)9 '©0 •i““é0“ù\$ •™ "Ù 'y#"ù\$" / '©M“ 6"Ù(21 'Y> 'é>', "Ù “és 'Y@'É?•ó
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #21.

- A) $\bullet Y^5 Y^2$, $\therefore \acute{e} = 0$ / Derivative value = 0
- B) $\text{"Y?"iM"}G\text{"y"}\acute{e}\$'\grave{U}.R\text{"9"}\grave{O} \grave{o} \text{æ} Ç\text{—}6\hat{A}6\ddot{o}ÇWF\text{—}\ddot{o}\hat{a}\grave{O}$
- C) $\text{"%."}\acute{e}\text{"")(\text{"%@"}\acute{e} = \pi / 2$ / Integral limit = $\pi / 2$
- D) $\acute{e}'>\text{©}>$ = 2 / Modulus = $\sqrt{2}$

Q.441. 'Y2', „6 Æ7VÇW2'Â 8i?"b , " 'y#''ù\$ •Y%''Y> •i5''Ù'')9 '©0 •i''é0''ù\$ •™ 'ù 'y#''ù\$' / '©M'' 6''Ù(22 'Y> 'és', 'ù 'és 'Y@'É?•ó
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #22.

- A) $\bullet Y^5Y^2$, $\cdot \acute{e} = 0$ / Derivative value = 0
B) $\text{"Y?"iM"}G\text{"y#"}\acute{e}\text{"}\grave{U}\cdot R\text{"}9^{\circ}\grave{O}\quad \grave{o}\quad \text{æ}\quad \text{Ç}—F-6\hat{A}\hat{6}\text{öÇWF}-\text{öä}\hat{O}$
C) $\text{"%."é"}(\text{"%@\acute{e}} = \pi / 2$ / Integral limit = $\pi / 2$
D) $\acute{e} > \text{©} > \text{"}$ / Modulus = $\sqrt{2}$

Q.442. 'Y2', „6 Æ7VÇW2'Â 8'i?"b , " 'y#""\$ •Y%"Y> •i5"Û"/)9 '©0 •i""é0"ù\$ •™ "Ù 'y#""\$ " / '©M" 6"Ù(23 'Y> 'é>', "Ù "és 'Y@'É?•ó
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #23.

- A) •Y5'Y2', .é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#ú\$Š"Ù. R 9"ñ Ò ò æ Ç—F-6 Â 6öÇWF-ðà Ò
C) "%..é")("%@'é>= pi / 2 / Integral limit = pi / 2
D) 'é>'©>) = " 2 / Modulus = sqrt(2)

Q.443. 'Y2', „6 Æ7VÇW2'Â 8i?'b ,” 'y#“ù\$ •Y%“Y> •i5'Ù'”)9 '©0 •i““é0“ù\$ •™ 'Ù 'y#“ù\$” / '©M“ 6'Ù(24 'Y> 'é>', "Ù “és 'Y@'É?•ó
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x ,
 determine the value / derivative / integral satisfying condition #24.

- A) •Y5'Y2', 'é = 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$""Û.'R 9"" Ò ò æ Ç—F—6 Â6öÇWF—öâ Ò
C) "%."é "(" ("%@'é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>)" = " 2 / Modulus = sqrt(2)

Q.444. 'Y2', „6 Æ7VÇW2'Â 8i?"b , " 'y#"ù\$ •Y%“Y> •i5"Ù"/)9 '©0 •i“é0“ù\$ •™ "Ù 'y#"ù\$ " / '©M“ 6"Ù(25 'Y> 'é>', "Ù “és 'Y@'É?•ó
 [English]: Evaluate the mathematical expression in Calculus / Vector Algebra / Trigonometry / Matrices: For real values of x, determine the value / derivative / integral satisfying condition #25.

- A) •Y5'Y2', •é(= 0 / Derivative value = 0
B) "Y?"iM")G"y#""é\$".Ù.'R 9"" Ò ò æ Ç—F-6 Â6öÇWF-öâ Ò
C) "%."é "(" ("%@'é> = pi / 2 / Integral limit = pi / 2
D) 'é>'©>)" = " 2 / Modulus = sqrt(2)